Relaxing the steady-state assumption displaces forest soil carbon models along the trade-off between litter input and turnover

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Draft of October 30, 2026

[Title: chosen before the storyline was rebuilt (trilemma, saturation); revisit. The verb must stay "displaces along", never "reveals"; no direction of turnover; Finland stays out.]

Abstract

[Abstract, to write last, as one arc of five to seven sentences. (1) The general problem: greenhouse gas inventories report the change of the soil carbon stock, most countries model that change, and the modelled change depends on the state the soil is assumed to start from; soil carbon carries a long history, and models started in balance with today's litter do not capture it. (2) The case: Finland, a forestry history that clearly changed, and three soil campaigns on the same plots, gaining and slowing. (3) The test: six models (three versions of Yasso and three simpler benchmarks), started in balance in 1985 or with the starting state estimated and dated back to 1917. (4) A balanced start loses carbon where the soil gained it; the estimated start reproduces the gain and moves the models along the trade-off between litter input and transit time. (5) Most of the modelled sink is inherited from before 1985, and fades: the sink is approaching saturation although the stock stays 21–40% below balance; about a tenth more litter input would postpone it. (6) The models reproduce the national trend but not the plots, whose residuals follow each stand's history.]

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[Table of contents: for the drafting phase only; remove before submission.]

1 Introduction

1.1 The general problem: a modelled soil sink depends on where the soil starts

1.1.1 Soil carbon change is usually modelled, not measured

1.1.1.1 OPENING Why greenhouse gas inventories matter

1.1.1.1.1 National greenhouse gas inventories are what shapes climate policy

Countries report their emissions and removals every year, and climate policies, national and EU, are evaluated against those reports. Accounted sinks or sources of greenhouse gases (GHG), at the national level, impact on national and EU policies, with direct economical and environmental impacts.

REFS: [to add to the bib:] UNFCCC Paris Agreement 2015; EU Regulation 2018/841 as amended by 2023/839

1.1.1.1.2 The soil holds most of the carbon and most of the uncertainty

Forests usually act as sinks: over 1990–2019 the world's forests took up about 3.5 PgC per year, close to half of fossil emissions (Pan et al., 2024). (ii) But this is not always the case: the sink of the boreal forests shrank by about a third over the same years, as disturbances grew (Pan et al., 2024). (iii) The boreal forests hold as much carbon per hectare as tropical forests (239 against 242 tC ha^{-1}; temperate forests 155) and about a third of all forest carbon (272 PgC), but most of it is in the soil, not in the trees: about 60% in the soil and 20% in the living biomass, where the tropical forests hold 56% in biomass and 32% in the soil (Pan et al., 2011). In boreal forests the forest carbon balance is therefore largely a soil balance. Because its C partially sits underground, the soil C balance is the least certain part of the total forest C balance. Also, high latitudes warm fastest, so the boreal soil is also where the climate pressure is largest.

REFS: Pan et al. (2024); Pan et al. (2011); [Pan et al. 2024: check the correction (Nature 632:E7) against the numbers; a source for the soil term as the largest uncertainty]; high-latitude warming: IPCC (2021) (land warms faster than the ocean, the Arctic at more than twice the global rate); Rantanen et al. (2022) (Arctic, north of 66.5°N, nearly four times the global rate over 1979–2021). [Both describe the Arctic, which the boreal zone only borders; a boreal-specific source would be tighter, e.g. Gauthier et al. 2015, Science 349:819–822, not yet read.]

1.1.1.2 OPENING What inventories focus on: the change, not the stock

1.1.1.2.1 The atmosphere sees only the change

The absolute amount of C stocks is not relevant for GHG balances, because stock that stays constant, however large, neither emits nor removes carbon; only its change exchanges carbon with the atmosphere. A national inventory therefore focuses on the annual change of the soil stocks, which enters the balance (next to other emissions, in the same units). The size of the stock is irrelevant for the balance, how it changes affects it sometimes dramatically.

REFS: [IPCC 2006 vol. 4 ch. 2 (stock-difference and gain–loss methods)]; Blujdea et al. (2015)

1.1.1.2.2 A small relative change can be a very large flux

A change of a few tenths of a per cent per year is small against the stock, but not against the emissions. Soils hold about 2400 billion tonnes of carbon to a depth of two metres, and fossil fuels release about 8.9 billion tonnes of carbon each year; a gain of four tenths of a per cent of the soil stock per year would therefore equal the whole of the fossil emissions (Minasny et al., 2017).

1.1.1.3 OPENING Most inventories relies on models

But such a small change in SOC is hard to detect, and measuring it requires expensive long-term campaigns that resample the same plot over decades. Most countries model it from litter input and climate.

1.1.1.3.1 Who measures, who models

Only a few countries measure the change by repeated soil sampling on inventory plots, Sweden (Swedish Environmental Protection Agency, 2026) and Germany (Grüneberg et al., 2014) among them; most model it. Finland, Norway, Switzerland and Austria run Yasso07 on their inventory plots (Statistics Finland, 2025; Norwegian Environment Agency, 2025; Didion et al., 2016). [

REFS: Swedish Environmental Protection Agency (2026); Grüneberg et al. (2014); Statistics Finland (2025); Norwegian Environment Agency (2025); Didion et al. (2016); Hernández et al. (2017); Jandl et al. (2014)

]

1.1.1.4 CHALLENGE A modelled change depends on many assumptions

But the results of a process-based SOC models depend on many assumptions. The most crucial is the initial state of the simulation, the more important the closer in time the start is to current state. That initial state is never measured. This effect dissipates over time, but can still persist for many decades after the start of the simulation and influences heavily the trajectories simulated by the model. The change a model reports is also the soil's response to its own past.

1.1.2 The initialization problem and why it matters: the soil remembers its past inputs

1.1.2.1 OPENING Two timescales

1.1.2.1.1 Carbon passes through in decades

The mean time a unit of carbon spends in the soil is usually decades (Sierra et al. 2017, 2018); 20–30 years in the models used here. But this does not mean that the initial state effects dissipates in such a short time.

REFS: Sierra et al. (2017); Sierra et al. (2018)

1.1.2.1.2 But he slow pools adjust over centuries

After a change of input we move from an equilibrium state to another and over this trajectory changes in C stocks occour. And the slowest C takes centuries to reach its new balance. The two timescales measure different things (Sierra et al., 2018). The mean transit time is an average over the carbon that enters the soil, and most of that carbon decomposes within a few years and leaves. The time a soil takes to reach a new equilibrium instead depends on where the carbon is stored, and most of the stored carbon sits in the slow components which keeps adjusting for centuries after a change of input (Wutzler and Reichstein, 2007). Most of the input leaves within years, so the average time carbon spends in the soil is decades; but most of the stock is built by the small share that accumulates.

1.1.2.2 CHALLENGE Today's soil reflects inputs of long before any model starts

Because of that, the soil keeps a memory of past land use and inputs far longer than its turnover suggests: a soil that underwent land use changes, or management changes, within the last century, is unlikely to be in equilibrium and the consequences of those changes are still present in current SOC stocks.

1.1.3 The usual assumption does not capture that history

Yet, SOC models are usually initialized assuming equilibrium conditions just a few decades before current state. This convention is justified with the obvious difficulties in reconstructing past history beyond that, but still is not going to represent reality.

1.1.3.1 OPENING Models are started in balance

1.1.3.1.1 Where the assumption is used

1.1.3.1.1.1 Inventories and models

The Finnish inventory, specifically, initializes Yasso07 assuming as constant the of inputs of the first available years in the time series (in the 70s) (Statistics Finland, 2025; Lehtonen et al., 2016).

REFS: Statistics Finland (2025); Lehtonen et al. (2016); Coleman and Jenkinson (1996); Kurz et al. (2009); Carvalhais et al. (2008); Luo et al. (2016)

1.1.3.2 CHALLENGE What a balanced start can and cannot carry

The assumption is that "approximately ten years of simulation since spin-up is enough to cancel out the effect of the spin-up level" Statistics Finland (2025), meaning that the years would be enough to offset eventual error carried out by the equilibrium assumption. The model still reacts to every change after its start date, so the assumption looks harmless. But with its published parameters and the mean climate and litter of our plots, Yasso07 started in balance differs from a soil that started 10 per cent below balance by 8.7 t C ha^{-1} at the start and still by 84 per cent of that difference after ten years. The soil would still be gaining 0.09 t C ha^{-1} yr^{-1} at that point.

Replacing the history of the site with an equilibrium with its present litter means that a soil that could still be recovering from past losses is set as if it had already recovered. This changes the trajectories that the model would simulate, since all changes in SOC stocks are given by a trajectory between two different equilibrium points (two land uses, or two C input levels). For a soil that is

gaining C, which is therefore clearly not in equilibrium, this assumption is particularly strong and forces us to explain the sinks with changes in C inputs to the soil happened after the start of the simulation. This is not impossible, but it is also quite likely that a soil that is a sink was not at equilibrium since already quite some time.

1.1.3.3 RESOLUTION A question raised before, from within the same community

That the assumption of a soil in balance at the start of a simulation is a coarse approximation that can lead to errors is nothing new and has been questioned already, often within the same community working around the Yasso model. Peltoniemi et al. (2004) found that the simulated carbon accumulation of 64 southern Finnish stands depended strongly on the initial stock, and that past slash-and-burn cultivation may have left soils well below balance. Peltoniemi et al. (2006) showed that the initial state was the largest source of uncertainty in the simulated soil sink at the start of their simulation. A review of seven models by Peltoniemi et al. (2007) stated that a balance with the current litter input "is likely to be violated in most applications", and Wutzler and Reichstein (2007) showed with Yasso that calibrating to stocks assumed to be in balance overestimates the decay rate of the slowest pool.

The closest antecedent of the present study is Liski et al. (2006), who ran Yasso through the litter input of Finnish forests from 1922 on in balance with the litter input of the 1920s. The present study follows the same route (and relies on that study for the shape of the litter input before the first measurements), but estimates the starting state of the soil from the soil carbon (now measured over three campaigns) on the same plots instead of assuming it.

1.2 The case: Finland, and the question of saturation

Finnish forests are a case in which the condition that a start in balance requires, a litter input that stayed constant for a long time before the start, is unlikely to hold. Centuries of slash-and-burn cultivation, forest grazing and selective cutting of the best trees left the forests of the early twentieth century sparse (Liski et al., 2006; Aakala et al., 2023; Korhonen et al., 2024). The war decades added a further drain: the war reparations of 1944–1952 were financed by the forest export economy (Mitrunen, 2024), the territory ceded in Karelia had held about 12 per cent of the forests (Kochetkova, 2020), and the use of wood in the 1950s and 1960s reduced the growing stock (Korhonen et al., 2021). Since then the growing stock has risen by 84 per cent between 1921 and 2023, mostly after the late 1960s (Figure 1). Because the litter falling to the soil is tied to the standing biomass, the litter input has risen over most of the century, and a soil with such a history is unlikely to be in balance with the litter input of any recent date.

▶ **Figure 1 here** (`F10b_forest_history`)

The soil itself was measured three times on the same permanent plots, in campaigns whose mean sampling years are 1989, 2006 and 2024. On the 310 plots measured in all three campaigns, the mean stock of the whole soil profile rose from 64.5 to 71.5 and to 73.6 $\mathrm{tC\,ha^{-1}}$, that is, by 0.40 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ until 2006 and by 0.12 afterwards; weighted by the area of forest that each plot represents, the two rates are 0.30 and 0.06 (Figure 2). The gain over the whole period is robust, but the interval of the later rate includes zero, so the measurements suggest, rather than show, that the gain is slowing. The national series is carried by the southern plots, which are 271 of the 310. In the south the stock rose by 0.48 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ to 2006 and by 0.16 afterwards, so the slowing is clearer there than

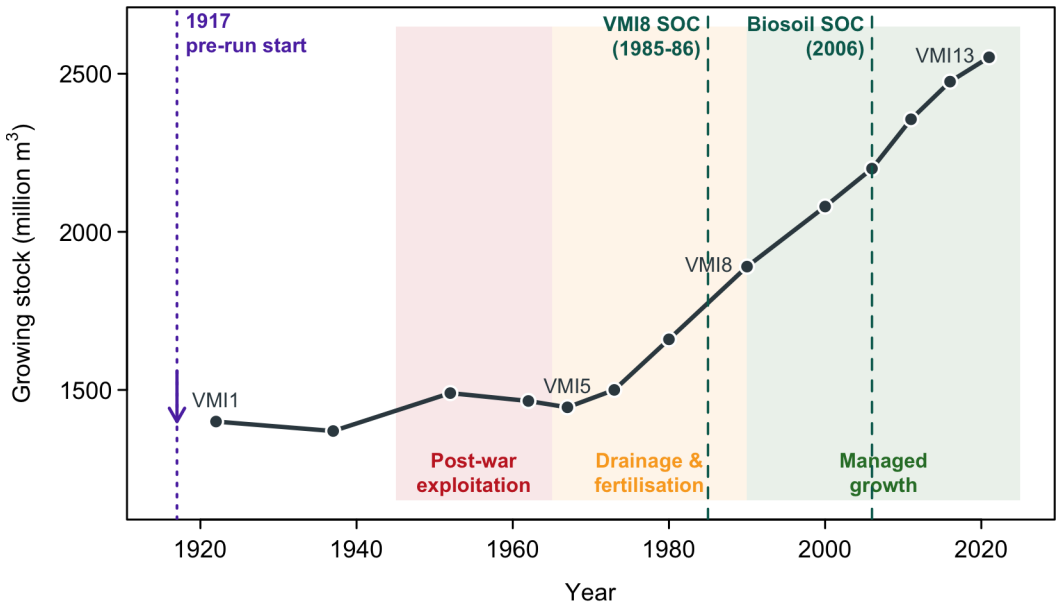


Figure 1: The growing stock of Finnish forests, 1921–2023. Growing stock rose by 84 per cent over the century shown, and most of the rise occurred after the late 1960s. The litter falling to the soil is tied to the standing biomass, so the litter input that the soil has integrated has been rising throughout the period that the models are asked to reproduce. Data from Korhonen et al. (2024).

in the national mean. The 39 northernmost plots, above about 66°N, lost carbon over the whole period, at 0.16 tC ha^{-1} yr^{-1} (95% interval 0.04–0.27), at the same pace before and after 2006, a result that rests on few plots and on how comparable the first campaign is in the north. None of the models reproduces the northern loss, and the conclusions of the paper therefore describe mainly the southern forests. ▷Supplement, Figure S2 (`S_soc_by_region`)

▶ **Figure 2 here** (`F3_mean_SOCalt`)

Whether the soil carbon sink of these forests is approaching saturation is the first question of the present study. Saturation is meant here as a sink that fades, not as a stock that reaches a ceiling. The question matters beyond the measured plots, because the national inventory reports the soil term from a model: the modelled mineral-soil term of the Finnish forests turned from a sink into a source from 2021 onwards, with a smaller litter input after the foliage estimates were recalculated given as one of the reasons (Statistics Finland, 2025), together with the warming climate (Natural Resources Institute Finland, 2025). In both directions, the reported term is the response of a modelled soil to a litter input that departs from the level at which the soil was started in balance.

1.3 The hypothesis: a trilemma

1.3.1 Three sources of information on the same ground

1.3.1.1 OPENING The three corners

1.3.1.1.1 The soil series

[Three campaigns on the same plots, 1985–2024, on one homogenised basis.]

REFS: Kramarenko (2012)

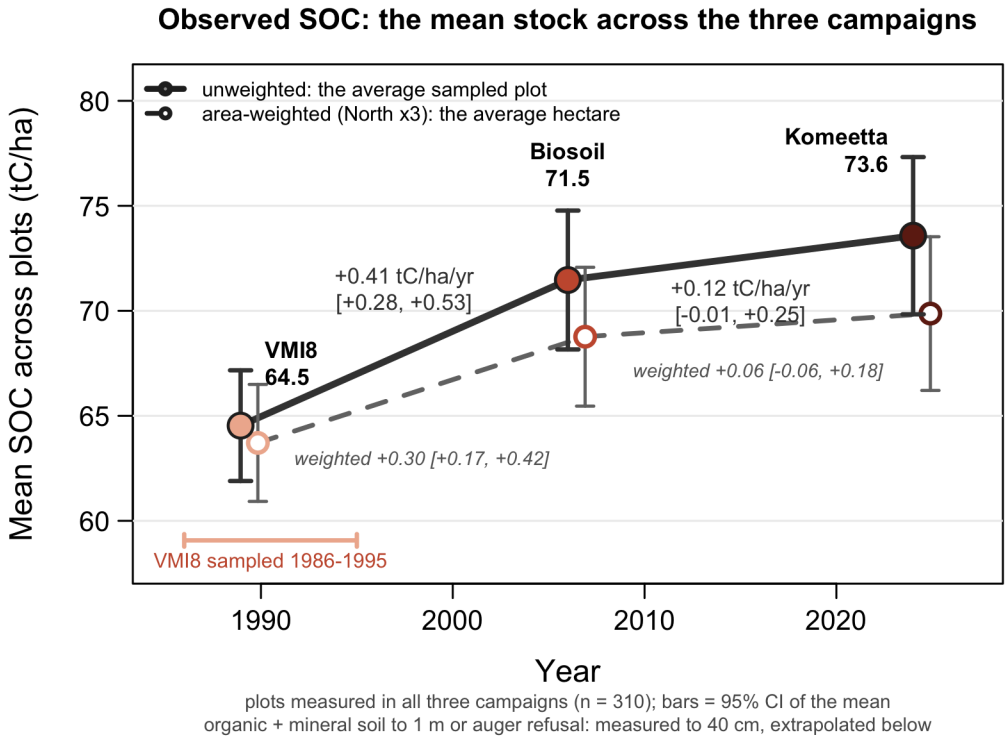


Figure 2: The measured soil carbon stock of Finnish forest plots in three campaigns. Each point is the mean stock across the plots measured in all three campaigns ($n = 310$), and the bars show the 95% confidence interval of that mean. The first campaign is drawn at the mean year in which its plots were actually sampled, and the bar low in the panel shows that the sampling of the first campaign ran from 1986 to 1995. Each segment is labelled with the rate of change over that interval and with the 95% confidence interval of the rate, calculated from the paired difference between campaigns on the same plots. The filled points and the solid line are the unweighted means, in which every plot counts equally; the unweighted means are the ones compared with the models. The open points and the dashed line are the means weighted by the area of forest that each plot represents, with a weight of three for the plots in the north, which are sampled at a third of the southern density. The stock is the whole soil profile: the organic layer and the mineral soil, measured to 40 cm and extrapolated below, down to one metre or to the depth at which the auger stopped. [Open: merge with Figure 1?]

1.3.1.1.2 The kinetics

[The published calibrations of the inventory's own model family (Yasso07, Yasso15, Yasso20): decomposition rates, transfers and climate response estimated on litter-bag and soil data worldwide.]

REFS: Tuomi et al. (2009); Tuomi et al. (2011); Viskari et al. (2022); YASSOmodel (2020)

1.3.1.1.3 The inputs

1.3.1.1.3.1 The litter record

[A litter record for every plot (Tupek et al.), from tree biomass models and turnover, rising by about 45% to a peak in 2006 and flat after; the north receives about two thirds of the southern input.]

REFS: Ťupek et al. (2026)

▶ **Figure 3 here** (`F1_inputs`)

1.3.1.1.3.2 The input corner is relatively robust

[On the scope shared with the national inventory (tree turnover and harvest residues), the product agrees with the inventory in the south, where 406 of the 456 plots lie, and neither rises after about 2006; the inventory also carries natural mortality, which the product lacks. ▷Supplement, Figure S3 (`S18_litter_vs_inventory`)]

[One clause on scope: robust for the tree litter it estimates; the understorey, natural mortality and below-ground inputs lie outside it. Plant, do not announce.]

REFS: Statistics Finland (2025)

1.3.2 Under the steady-state assumption, two of the three at most

1.3.2.1 CHALLENGE Why the three are tied

1.3.2.1.1 In balance, stock = input × transit time

[In a soil in balance the stock is the input times the mean transit time, so the three sources constrain one another: given two, the third is fixed. A soil that keeps gaining carbon while its litter has stopped rising cannot be in balance with that litter.]

REFS: Olson (1963); Sierra et al. (2017)

1.3.2.2 RESOLUTION The hypothesis

[Under the steady-state assumption the soil series, the kinetics and the inputs cannot be reconciled: a model started in balance can satisfy two of them at most.]

1.3.2.3 RESOLUTION Where it is tested, and where it comes back

1.3.2.3.1 Tested

[At the opening of the Results (R1), with the balanced-start calibrations, which cannot appear in the Introduction.]

1.3.2.3.2 Its threads

[The ridge between input and turnover (R2), the forecast (R3), the history of each plot (R4).]

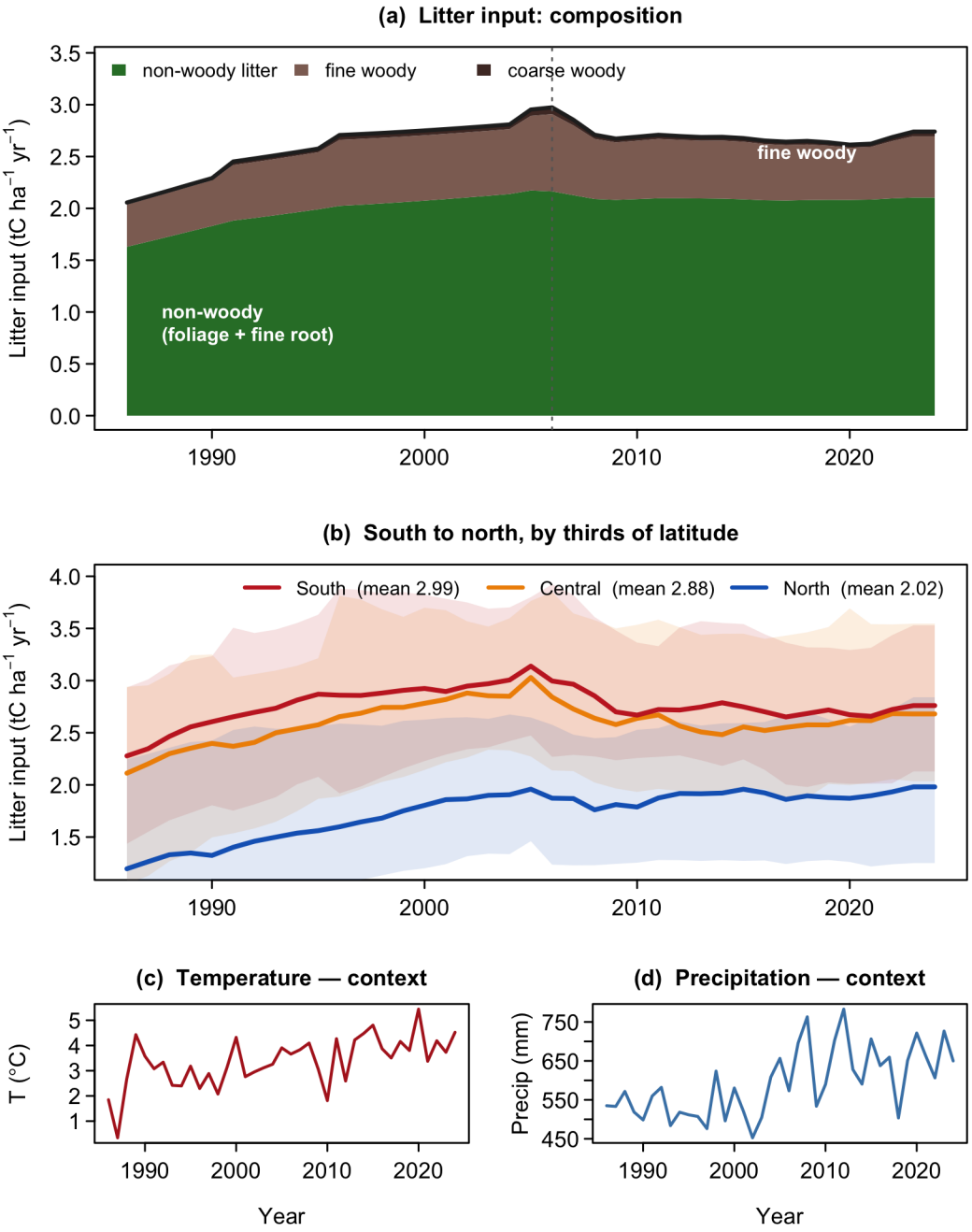


Figure 3: The litter input to the study plots, 1986–2024, from the plot-level product of Ťupek et al. (2026). (a) Composition of the mean input across plots: non-woody litter (foliage and fine roots), fine woody and coarse woody litter; the total rises from 2.1 to 3.0 tC ha^{-1} yr^{-1} at its 2006 peak and stands at 2.7 in 2024. (b) The total input by thirds of latitude, with the interquartile range across the plots of each third; the north receives about two thirds of the southern input. (c, d) Mean annual temperature and precipitation over the plots, for context.

1.4 Objectives

1.4.0.1 ACTION The test

[The same six models, on the same data, started in balance at the beginning of the measurements and with the starting state estimated from the measurements and dated back to 1917. Open: name the balanced-start comparison here, or only in the Methods?]

1.4.0.2 CHALLENGE Four questions

(i) Can a model started in balance reconcile the soil series, the kinetics and the inputs, and does relaxing the assumption?

(ii) Where does reconciling them place the models on the trade-off between litter input and transit time?

(iii) Under the relaxed assumption, is the forest soil sink approaching saturation, how does it respond to warming, and how much additional litter input would postpone saturation?

(iv) Why do the models reproduce the national trend but not the differences between plots?

1.4.0.3 RESOLUTION Scope of the applied answer

[The projections describe the calibrated models on the study plots, per hectare; the inventory's term is not re-estimated, and the baseline projection holds the litter at 2024 and repeats the recent climate. The answer to (iii) is about direction and sensitivity, not a forecast of the reported numbers. National units would need a sourced area and Aleksi's review.]

2 Materials and methods

The section first describes the data (the soil carbon observations, the litter input and the climate), then the six models and how they are started and calibrated, and then the calibrations and simulations built on that common setup: the comparison with a soil started in balance, the division of the modelled sink among its sources, and the simulations beyond the observations. The evaluation of the models against the observations closes the section.

2.1 Study plots and soil carbon observations

The study is based on data from permanent forest plots of the Finnish National Forest Inventory for which soil carbon has been measured in three campaigns: the eighth national forest inventory (VMI8), the BioSoil campaign of 2006, and the Komeetta campaign of 2024. After the exclusions described below, 456 plots are available for calibration, carrying 1205 plot-year observations of soil carbon in total. Of the 456 plots, 365 plots (80 per cent, carrying 959 observations) were used to fit the models, and 91 plots (20 per cent, carrying 246 observations) were held out and used only for evaluation. Plots were assigned to the two groups at random, and a plot was assigned as a whole, so that no plot contributes observations to both groups.

The first campaign is not a single year. Its plots were sampled region by region between 1986 and 1995: 69 plots in 1986, 76 in 1987, 103 in 1988, 54 in 1989 and 72 in 1995. Averaged over the plots, the sampling year of the first campaign is 1989, and that is the year at which the campaign is

drawn whenever it is shown as a single point. In the calibration, each observation is compared with the model in the year in which that plot was actually sampled.

On the plots measured in all three campaigns (310 plots, the set on which every observed rate of change in this paper is computed), the mean stock of the groups sampled in 1986, 1987, 1988, 1989 and 1995 differs by 18.3 tonnes of carbon per hectare at first visit, which might reflect a trend within that same campaign. The five groups of plots defined by their first-campaign sampling year differ by 37.2 tonnes in 2006 and by 34.2 tonnes in 2024, years in which every plot of every group was measured in the same campaign. The difference between the five groups is therefore a property of which plots each group contains, and not of the year in which it was sampled, because the difference is larger when the sampling year is held constant.

[The exclusion rules (zero litter, absent organic layer, peat soils, constant litter series, whole-profile stock above 250 t C ha^{-1}, and the 82 first-campaign observations without a recorded sampling year) are referred to as "described below" but are not yet written out; add one short paragraph listing them with the number of plots each removes.]

2.2 Soil carbon stocks on a single homogenised basis

Since the three campaigns were measured and processed under protocols that differed in ways that matter for a comparison between them, all three campaigns were rebuilt on a uniform basis, starting from the layer-by-layer carbon stocks measured in each of the three campaigns and applying one set of conventions to all three. Those stocks are corrected for the volume occupied by stones and for the bulk density of the fine earth. The volume of stones was measured in 2006 on every plot by the rod-penetration method of Viro (1952), in which a steel rod is driven into the mineral soil at the sampling points and the mean depth it reaches is converted into a stone volume, and the same stone volume is applied to the first campaign (Lehtonen et al., 2016; Kramarenko, 2012). The bulk density of the mineral layers was predicted from the equations of Tamminen and Starr (1994) for all layers of the first campaign and for most layers of 2006, and measured directly for part of the 2006 samples from the upper 20 cm (Kramarenko, 2012). [Confirm that the 2024 campaign uses the same stone volumes and the same bulk density equations.] The coarse-fragment correction matters because the coarse-fragment content of these plots has a median of 43 per cent by volume, so a mineral stock computed without it is too large by a factor of about 1.6.

The quantity used as the calibration target is the whole soil profile: the measured organic layer, plus the measured mineral soil to 40 cm, plus a modelled deep tail, integrated only as far down as the auger actually reached on that plot and in no case deeper than one metre. The depth to which the profile is integrated is therefore a per-plot quantity and not a fixed one metre, since using a fixed depth of, for example, one metre would invent soil carbon below bedrock on the 38 plots where augering stopped at 10, 20 or 40 cm. Below 40 cm the stock was extrapolated assuming that the carbon content decreases exponentially with depth. One rate of decrease was fitted for each soil class of the national soil map, jointly on the profiles of all three campaigns, and the level of each profile's curve was set by that profile's own measured mineral layers. The first campaign was sampled in different depth intervals (0–5, 5–20 and 20–40 cm, against 0–10, 10–20 and 20–40 cm in the two later campaigns); the fit uses the actual interval of each layer, so the two layer schemes need no conversion. The depth reached by the auger is recorded only in 2006 and, like the stone volume, is applied to the first campaign. In the first campaign the carbon content of the mineral layers below 5 cm was mostly predicted from the loss of mass on ignition rather than measured

(Kramarenko, 2012), and because the deep tail is scaled from those layers, the deep tail carries the same uncertainty. The extrapolated part is about 15 per cent of the profile stock in each of the three campaigns. The modelled deep tail was validated against the measured 40–80 cm layer of the BioSoil campaign, which was held out of the fit: the median ratio of predicted to observed carbon in that layer is 0.96, with a bias of −2.5 tonnes of carbon per hectare.

The rebuilt target was checked against the national figures published for the two campaigns for which such figures exist. Applying the same plot exclusions as the official calculation, and integrating over the same layers, the rebuilt target returns weighted national means of 59.06 and 61.05 tonnes of carbon per hectare for 2006 and 2024 against official values of 59.06 and 61.05, that is, agreement to the nearest 0.01 tonnes per hectare on 446 plots.

Two further steps were taken with the first campaign, both of them established from the structure of the measurements themselves rather than from the fit. First, the layers of the three campaigns were standardised, because they were recorded differently: the organic layer of the first campaign contains the fermentation and humus horizons only, whereas the organic layer of the two later campaigns also contains the litter and moss layer. Each plot's own litter and moss layer from 2006 was therefore added to its first-campaign organic layer (for the 20 calibrated plots without a value from 2006, the median of all plots), so that the same compartment is compared across the three campaigns. Second, the first campaign is treated as what it is, a campaign whose fieldwork ran over several years rather than a campaign carried out in a single year. The year in which each plot was actually sampled is on record, and those years, rather than one nominal year for the whole campaign, are the years at which the models are compared with the observations.

On this final basis, over the 1205 observations used in this study, the median whole-profile stock is 61.4 tonnes of carbon per hectare for the first campaign, 65.3 for 2006 and 66.8 for 2024, and the observations vary between plots with a coefficient of variation of 0.36 to 0.48 depending on the campaign. (Figure S1) [wrong figure: Figure S1 shows the uncalibrated Yasso20, not the soil carbon basis; drop the pointer or replace the figure]

[Data availability and acknowledgement: the soil carbon measurements are not a published dataset, so the statement has to say where they come from, who compiled them and what of the compiled form can be deposited. Settle this with the people who hold them before submission.]

2.3 Litter input

Litter input is taken from the annual reconstruction of tree biomass and litter input for the permanent plots of the Finnish National Forest Inventory by Ťupek et al. (2026), published as an open dataset. In that reconstruction, the tree biomass measured by the inventory is interpolated to every year with the help of satellite-derived basal area, and litter is computed as the biomass of each tree component multiplied by a turnover rate for that component, plus the branches, foliage, stumps and roots left on site when a drop in basal area identifies a harvest; the full method is given in Ťupek et al. (2026). The product supplies, for each plot and year, the carbon in litter from trees, separated into non-woody litter (foliage and fine roots), fine woody litter (branches and coarse roots) and coarse woody litter (stumps), and each of the three classes is further divided into the four chemical fractions used by the Yasso models, using proportions specific to each tree species and component. The product does not contain a separate term for trees that die without being harvested, and it does not include the understorey vegetation, because the product is a tree litter product by design. Across the study plots the mean litter input carried by the product is $\bar{J} = 2.511$ tonnes of carbon per hectare per

year.

The national greenhouse gas inventory estimates the same flux for the mineral forest soils of southern and northern Finland, from the forest inventory and biomass models (Statistics Finland, 2025). On the part that the two estimates have in common, the litter from the turnover of living trees and from harvest residues, they agree in level in southern Finland, where most of the study plots lie, and neither rises after about 2006; in northern Finland the inventory estimate lies higher and keeps rising ▷ Supplement, Figure S3 (`S18_litter_vs_inventory`). The inventory estimate also carries the litter of trees that die without being harvested, a term that grows over the period and that the plot product does not contain. [confirm with LUKE that the two regions of the inventory data match the region code of the study plots]

The first year of the product is treated separately. The median litter input of 1985 is 0.060 tonnes of carbon per hectare per year against 1.398 in 1986, a step that the series does not repeat in any later year. Because 1985 is both the first year of the simulation and the year in which the first soil observations fall, the value for 1985 was reconstructed for each plot and each chemical fraction by fitting a linear trend over 1986–1990 and extrapolating one year backwards. The reconstruction places 1985 just below 1986, which is the direction the growing-stock record requires.

2.4 Climate

Climate forcing comes from the gridded daily weather observations of the Finnish Meteorological Institute, interpolated from the station network onto a 10 by 10 km grid by kriging with external drift (Aalto et al., 2016) and extended by the Institute to the present. The grids were obtained through the internal weather database of the Natural Resources Institute Finland, which mirrors them from the Finnish Meteorological Institute. From the grid cell nearest to each plot we took the daily mean, minimum and maximum temperature, precipitation and global radiation for 1961–2025, and each model takes from those series the variables it requires. The record is used in two ways. From 1985 onwards each year of the simulation is driven by the climate observed in that year. Before 1985 the simulation is driven by one constant climate, the mean of 1961–1984 at each plot (a plot mean of 2.2 degrees Celsius, against 2.7 for 1985–2004), so that the soil enters 1985 adjusted to the climate it had lived in; the same long-term mean serves as the reference climate wherever one is needed later (Sections 2.7, 2.13 and 2.14).

Because the response of decomposition to climate is not a straight line, the average of the response over the years is not the response evaluated at the average climate. Averaging the climate first and evaluating the response afterwards biases the climate modifier by 5 to 15 per cent in boreal conditions. The climate modifier is therefore computed for each year and averaged afterwards, in all six models.

2.5 The six models

Six models are used (three Yasso versions, plus three simpler benchmark models), and the two groups have different roles. Yasso is the model this study is focused on, because Yasso is the model the Finnish greenhouse gas inventory is currently based upon. All three published versions of Yasso are included: Yasso07, Yasso15 and Yasso20. Alongside the three versions of Yasso we included three simple models with one, two and three carbon pools, denoted SP1, TP2 and TP3. The three simple models are there as a benchmark, since their behaviour is transparent enough so that any result

obtained with Yasso can be checked against a structure simple enough to reason about quickly.

Due to their different structures and designs, the number of free parameters differs greatly across the six models, from six free parameters in SP1 to twenty-six in Yasso15 and Yasso20. The free parameters of the three versions of Yasso are the twelve fractions describing transfers between pools, the climate parameters, the parameters describing the effect of the size of woody litter, and the two input parameters described in Section 2.6.

The decomposition rates of Yasso are not fitted, and are held at their published values. Holding the rates fixed is a deliberate choice to keep the model anchored to its original definitions. Since decomposition rates and transfer fractions do not act independently on the carbon stock but are always interacting, a slower rate and a larger transfer into a slow pool produce very similar trajectories, so fitting both together adds little information and makes the result harder to read.

Whether the pools of a soil carbon model can be identified with fractions of organic matter isolated in the laboratory is a long-standing question in soil carbon modelling, and has been examined at length (von Lützow et al., 2007). A chemical fractionation is itself a discretization: it divides a continuum into a small number of classes, and where the boundaries between classes fall is set by the method chosen, so that the fractions are defined by the procedure that isolates them (von Lützow et al., 2007). The material being divided is not necessarily a set of discrete compounds, since the chemical diversity of soil organic matter is large enough that the diversity can be treated as a continuum (Lehmann and Kleber, 2015). Yasso was built on the assumption that the pools can be identified chemically, and that litter consists of groups of organic compounds, each group decomposing at its own rate regardless of the litter it came from, and defined its decomposition compartments accordingly as extractives, celluloses and lignin-like compounds, followed by two compartments of humus (Liski et al., 2005). All following Yasso versions are built on the same assumption, and the models have been calibrated on data derived from such assumption, which is therefore a fundamental part of the model. The same aim, of matching the pools of a model with fractions that can be isolated in the laboratory, has been pursued for other models, for example through fractionation schemes designed to correspond to the pools of RothC (Zimmermann et al., 2007). Whatever the pools are meant to represent, however, and whatever the nature of soil organic matter, which remains debated, a compartmental model defines each pool by its decomposition rate: the rate groups carbon by how long it stays in that pool, its turnover time (Sierra et al., 2017), and changing a rate changes what the pool is.

We therefore treated the rates as part of the definition of each model, kept the rates as the published parameterisation set them, and calibrated the transfer fractions. In all three versions of Yasso the decomposition rates, together with the fraction of decomposed carbon that enters the humus pool, are fixed at their published values: they are constants of the model and not parameters of the calibration, and no prior is placed on them.

2.6 The uncertainty of the litter input

The litter input is among the least certain of the quantities that drive a soil carbon model. Litter is not measured on the plots but derived from tree biomass. Parts of the input, such as the litter of the understorey, the litter of trees that die naturally and root exudates, are missing from the product used here (Section 2.3), and the input carries year-to-year variation that periodic inventories cannot capture. Uncertainty analyses of the Finnish soil carbon estimate have already identified the litter input, together with the initial state of the soil, as the dominant sources of uncertainty (Peltoniemi

et al., 2006; Lehtonen and Heikkinen, 2016). Those analyses propagated the uncertainty forward by Monte Carlo simulation, sampling the input, and in Lehtonen and Heikkinen (2016) also the parameters of Yasso07, from error distributions derived from the data sources, and running the soil model with each sample. We instead estimate the error of the input from the soil observations themselves, together with the parameters of the models, so that the correction applied to the input and the uncertainty of that correction are both learned from the data and carried into every result.

The correction is a single multiplier, σ_{input}, the *input multiplier*, applied to the litter input of the product over the whole simulation. The prior of σ_{input} is centred on 1.08, the ratio of the total litter input to Finnish upland soils estimated by Lehtonen and Heikkinen (2016), which includes the understorey, to the tree litter input of the product used here, 2.70 against 2.511 tonnes of carbon per hectare per year. The prior is set on the litter input that σ_{input} produces, which is kept within 0.05 to 8.7 tonnes of carbon per hectare per year, the upper end of the net primary production reported for evergreen forests (Gower et al., 2001), so that no value of σ_{input} can produce an impossible input. Within that range the prior gives σ_{input} a median of 1.07 and a 95 per cent probability of lying between 0.75 and 1.45. Because the multiplier is one number for the whole country, it rescales only the national level of the input and not the input of individual plots, whose errors are carried by the error model described below.

The same reasoning is extended to the initial state. The litter input of 1917, which sets the state of the soil at the start of the simulation, is not known either, and is estimated through a second parameter, σ_{init}, the *initial input ratio*, the ratio of the litter input of 1917 to the litter input of 1985. The prior of σ_{init} is centred on 0.90, the ratio of the growing stock of Finnish forests in 1917 to that in 1985 (Korhonen et al., 2024) converted to a ratio of litter input with the relation between litter input and growing stock observed in the product, and is constructed in the same way, on the litter input of 1917, giving a median of 0.93 and a 95 per cent range of 0.58 to 1.49; the reconstruction of Liski et al. (2006), based on the same growing-stock record, implies a similar ratio, about 0.86. Both parameters are estimated together with the model parameters, so that the uncertainty of the input and of the starting state propagates into the initial state, into the simulated trajectory and into the predictive intervals of every model.

2.7 Transient initialisation

Rather than starting each plot from a soil in balance with its present litter input, each draw of the parameter vector starts the soil in 1917 and runs it forward to the first soil observation before the comparison with data begins. The year 1917 is a convention, chosen as the year of Finnish independence. The state of the soil in 1917 is obtained conventionally, with a steady state assumption with a constant litter input equal to the estimated litter input of 1917 and constant climate (calculated as the mean of 1961–1984, Section 2.4). The litter input is then raised along a prescribed shape over the sixty-eight years from 1917 to 1985, still under the same constant climate, after which the simulation is driven by the annual litter and climate of the record. A constant climate is used before 1985 because the records start in 1961 and do not reach back to 1917; the mean of the earliest observed decades is the closest available measure of the climate in which the soil developed before the observations began.

The shape along which the litter input is raised is taken from the reconstruction of the litter input to Finland's forest soils published by Liski et al. (2006), who estimated that input annually from 1922 onwards by combining national forest inventory data with biomass and turnover models.

Three decisions were made about that reconstruction.

First, only the tree-derived terms are used, meaning tree litter together with harvest residues (and natural tree mortality), because those are the terms closest to the litter product used after 1985. The plot-level product of Ťupek et al. (2026) has no term for natural mortality: a tree that dies is registered only as a loss of basal area, and its stem does not enter the soil. Natural mortality is, however, only 2 to 4 per cent of the tree-derived input in Liski et al. (2006), and leaving it out of the shape changes no year by more than 0.04 on the scale of zero to one defined below. The ground vegetation from Liski et al. (2006) is left out for uniformity with the product used after 1985, assuming that it is absorbed by the input multiplier described in Section 2.6. Including it before 1985 but not after would make that multiplier mean two different things on either side of the join.

Second, the national totals are converted to a per-hectare basis by dividing by the area of upland forest soil, derived from Liski et al. (2006). Liski et al. (2006) report the carbon of upland soils both as a national total and per unit area, and the ratio of the two gives an upland soil area of 13.90 million hectares in 1922 and 15.22 million hectares in 2004. The area is held at the first value until 1965, increases linearly to the second over 1965–1980, the period in which Korhonen et al. (2024) locate most of the expansion of productive forest land, and is held constant afterwards.

Third, only the shape is taken from Liski et al. (2006) and not the level: the series is rescaled to run from zero in 1917 to one in 1985, so that the reconstruction supplies only the path between the two endpoints. The litter input in year t of the pre-run is $J(t) = J_{1917} + (J_{1985} - J_{1917}) \times s(t)$, where $s(t)$ is the rescaled shape; the same weighting is applied to each chemical fraction and size class of the input, so that the composition of the input is that of the plot in 1985–1989 throughout and only its amount changes. The two endpoints (J_{1917} and J_{1985}) are set by the two input parameters described in Section 2.6: the input at the end of the path is the litter input of the product, averaged over 1985–1989 for each plot and scaled by σ_{input}, and the input at the start of the path is a fraction σ_{init} of the input at the end.

► **Figure 4 here** (`F4_initialization_by_region`)

2.8 Keeping the models as comparable as possible

The six models cannot be given the same amount of external information. The models differ in structure and in the number of their parameters, and the published literature provides different information for each of them: the three versions of Yasso come with their own calibrations, whereas the three simple models have no calibration for forest soils. The more the models differ in the information they are given, in their assumptions and in their freedom, the less a difference between their results can be attributed to their structure. We therefore reduced that difference as far as the models allow, through one compromise for each class of parameter, summarised in Table 1. The compromises reduce the asymmetry between the models without removing it, and this must be considered when interpreting the results.

2.8.1 Decomposition rates

The rates of the three versions of Yasso are fixed at their published values, for the reasons given in Section 2.5. The three simple models have no rates of their own for forest soils, and take theirs from the ICBM model (Andrén and Kätterer, 1997): the fast rate is fixed, and the slow rate or

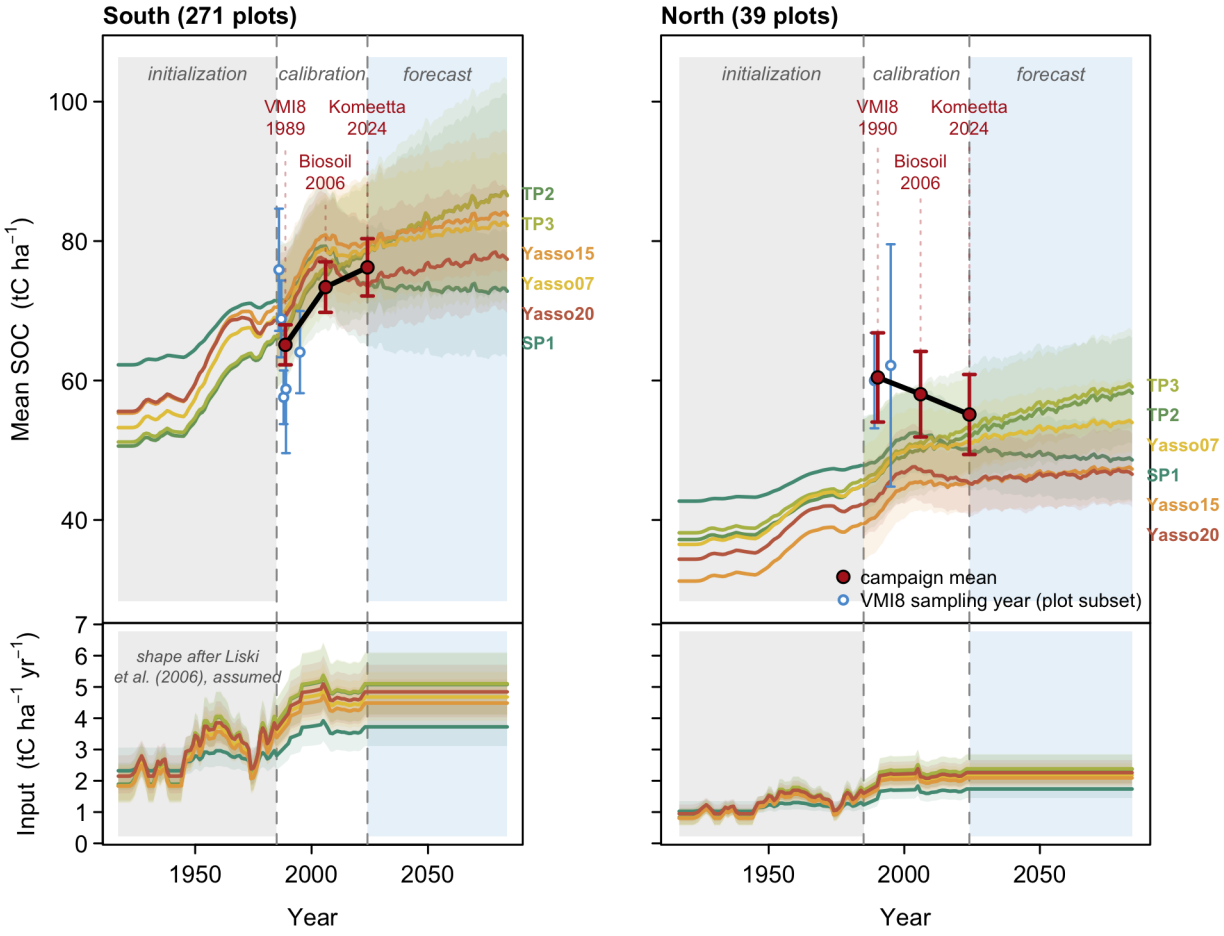


Figure 4: How the starting state of the soil is obtained. Rather than starting each plot from a soil already in balance with its present litter input, the simulation starts in 1917 from a soil in balance with the litter input of 1917 and the mean climate of 1961–1984, and the litter input is then raised over the following sixty-eight years along a shape taken from the reconstruction of Finland's forest litter input by Liski et al. (2006), after which the simulation is driven by the annual litter and climate of the record. The litter input of 1917 is not known and is estimated as part of the calibration. The left column shows the 271 southern plots and the right column the 39 northern plots measured in all three campaigns; in each column the upper panel is the mean soil carbon stock and the lower panel the mean litter input that drives it.

rates are given a very informative prior centred on the ICBM value, so that the data can move them only slightly. The slow rates are pinned rather than fixed because the ICBM values cannot be transferred to a forest soil unchanged. ICBM was calibrated on an arable long-term experiment, and cropland and forest soils differ in what controls decomposition. Moreover, a rate is only meaningful together with the climate response it was estimated with, and the simple models combine the rates of ICBM with the climate response of Yasso07; the rates of ICBM were divided once by the value of the Yasso07 climate modifier calculated on the climate of the experiment on which ICBM was calibrated (0.94). Fitting the rates freely (or under wide priors), on the other hand, would give the simple models a freedom that the versions of Yasso do not have, and a narrow prior around the ICBM value is the compromise between the two.

2.8.2 Transfer fractions

The fractions that describe how carbon passes between pools are free in every model that has them: twelve in each version of Yasso, and the fraction of decomposed carbon passed on to the slower pool in the simple models, of which SP1 has none, TP2 one and TP3 two. All fractions carry a prior of the same width, 0.4 on the logit scale, centred on the published value of each model, which for the simple models is the humification coefficient of ICBM.

2.8.3 Climate response and size of woody litter

The three versions of Yasso take the centres and widths of their climate priors from their own calibrations: Yasso07 from Tuomi et al. (2009, 2011); Yasso20 from the sample of 10 000 parameter sets of Viskari et al. (2022), distributed with the model code (YASSOmodel, 2022); and Yasso15, whose calibration has not been published in a journal, from the posterior sample distributed with its model code (YASSOmodel, 2020). The simple models, which have no published climate response of their own, borrow the climate response of Yasso07, centre and width. Only the versions of Yasso describe how decomposition depends on the size of woody litter, with priors taken from their own calibrations; the simple models have no such term.

2.8.4 The two input parameters

Every model carries the two parameters described in Section 2.6, the input multiplier σ_{input} and the initial input ratio σ_{init}, with the same prior in every model.

Table 1: How the prior of each class of parameter was set.

[Parameter-class by prior-criteria table: for each class (decomposition rates, transfer fractions, climate response, size of woody litter, the two input parameters), the source of the centre, the source of the width, the transformation and constraint, and which models carry it. The worked version is in `Prior_specs/PRIOR_HOMOGENIZATION_PLAN.md`, section 2.]

2.9 Numerical integration

Each model is run forward one year at a time, from 1917 to the last observation, with the litter input and the climate of each year held constant within that year. Within each year the equations of every model are solved exactly rather than stepped forward. For the one-, two- and three-pool models,

in which carbon moves in one direction only, the exact solution has a closed form, an exponential approach of each pool to its balance value, and is evaluated directly. In the three versions of Yasso carbon moves between pools in every direction, and the solution, the exponential of the matrix of rates and transfers, is computed in the Fortran code of the models by scaling and squaring (Moler and Van Loan, 2003). All models are run with an annual time step, the step for which Yasso was designed and calibrated; the seasonal course of temperature enters through the annual temperature amplitude in the climate modifier (Tuomi et al., 2009), which the simple models share.

2.10 Mean transit time as a common measure of a model

The six models differ in the number of their pools and in how carbon moves between those pools, so their parameters cannot be compared directly. The models are compared instead through one number, the mean time a unit of carbon spends in the soil between entering it as litter and leaving it as carbon dioxide, which Sierra et al. (2017) call the mean transit time. The same quantity is often called the mean residence time; Sierra and coauthors show that the term residence time has been used for at least three different quantities, and we follow their convention. For a model whose pools lose carbon in proportion to the carbon they hold, and whose rates do not change with time, the mean transit time equals the carbon the system holds at steady state divided by the carbon entering it (Rasmussen et al., 2016; Sierra et al., 2017). Each model is therefore given an input of one tonne of carbon per hectare per year, and its steady-state stock is computed directly: as a formula for the three simple models, and as the solution of one linear system for the three versions of Yasso. Because the models are linear, the ratio of stock to input does not depend on the input chosen.

The mean transit time computed in this way is a property of the model alone. Neither σ_{input}, nor σ_{init}, nor any observation enters the calculation, so different models (and the calibrated and the published parameterisations of the same model) are compared on the same scale. Because the models respond to climate and to the quality of the litter, the mean transit time for every model and every parameter set is evaluated at one reference condition: the mean climate of the study plots over all the observed years (mean annual temperature 3.4 °C, temperature amplitude 13.8 °C, annual precipitation 591 mm), and the mean litter of the study plots, divided into 0.766 non-woody, 0.219 fine woody and 0.014 coarse woody litter, each with its own chemical composition. The values reported are therefore those of each model under that condition, not those of the real soil under changing conditions; the comparison between parameterisations does not depend on the reference, because both sides are evaluated at the same one.

2.11 The error model

Observations are compared with predictions on the logarithmic scale, so that the error is proportional to the size of the stock. The total standard deviation of the error is fixed at 0.800 on that scale, about twice the spread of the observed stocks around the national mean of each campaign, 0.39. That spread is the error of an idealised model that follows the national mean exactly and knows nothing about individual plots. A process-based model adds information that the measurements do not contain, in the litter input, the climate and the representation of decomposition, and is expected to depart from the measurements where the measurements are in error. The realistic error is larger still, for four reasons: the litter input carries errors of its own at each plot, which counted as an independent error alone raise the total to 0.57; the observed errors have heavier tails than a normal

distribution; outlying plots were excluded from the sample; and part of the measurement error, in the stone volume, the bulk density and the modelled deep tail of the profile, is shared between plots and does not appear in a spread between plots. A value at the upper end of the plausible range was chosen, because a larger error weakens the pull of the observations against the priors and keeps conclusions drawn from the observations conservative.

Observations of soil carbon are not independent of one another: observations of the same plot, of the same part of the country and of the same campaign share whatever the models get wrong about that plot, that region or that protocol. The error of each observation is therefore divided into four parts,

$$\log y_{ij} \;=\; \log f_{ij}(\theta) \;+\; u^{R}_{r(i)} \;+\; u^{P}_{i} \;+\; u^{C}_{j} \;+\; e_{ij},$$

where y_{ij} is the stock observed on plot i in campaign j, $f_{ij}(\theta)$ the stock predicted by the model, u^R a part shared by all plots in the same band of latitude, u^P a part shared by all observations of the same plot, u^C a part shared by all observations of the same campaign, and e a part that belongs to the observation alone. Each part has a mean of zero and a fixed standard deviation, and no part is estimated for any plot, band or campaign: the likelihood is averaged over all the values the shared parts could take, which for normal errors can be done exactly and leaves only a correlation between observations that share a plot, a band or a campaign.

The standard deviations of the shared parts were taken from the observations after removing the mean of each campaign: 0.117 for the regional part, the spread between the means of eight bands of latitude holding equal numbers of plots, and 0.396 for the plot part, from how strongly two observations of the same plot vary together. The part that belongs to each observation alone, 0.685, is what remains of the total once the shared parts are accounted for, the variances (the squares of the standard deviations) of the four parts adding up to 0.800^2. Dividing the error does not change how far a single observation may lie from the model, but observations that share part of their error partly repeat one another, and the 1205 observations carry about as much information on the national mean as 220 independent observations would. Both published calibrations of Yasso treated the observations as independent (Tuomi et al., 2009; Viskari et al., 2022), the second naming the correlation of errors within a time series as an effect still to be included, and treating correlated observations as independent makes parameter uncertainties too narrow (Oberpriller et al., 2021).

The standard deviations of the campaign parts cannot be measured, because a shared offset between campaigns and a genuine change of the stock look the same in the observations, and were therefore prescribed: 0.030 for each of the two later campaigns, which followed similar protocols, and 0.060 for the first, which differs from them in three documented ways. Its organic layer excluded the litter and moss layer, added here from each plot's measurement of 2006; its fieldwork extended over a decade; and the carbon content of its mineral layers below 5 cm was mostly predicted from the loss of mass on ignition rather than measured (Kramarenko, 2012).

2.12 Sampling

Each model was calibrated with five independent runs of 50 000 iterations, of which the first 5 000 were discarded, with the differential evolution Markov chain sampler of ter Braak and Vrugt (2008), which adapts its proposals to strongly correlated posteriors, as implemented in the `BayesianTools` package (Hartig et al., 2023), version 0.1.9, under R 4.6.1 (R Core Team, 2023). Each run maintains three interacting chains, so that the five runs together provide fifteen chains, and every calibration

in this paper uses the same settings. Convergence was assessed with the potential scale reduction factor and the effective sample size: over all eighteen calibrations (six models, each with the transient initialisation, with a balanced start, and with a balanced start and the climate held) the largest potential scale reduction factor of any parameter was 1.009 and the smallest effective sample size 1716.

The information that the observations add to each parameter was measured as the Kullback–Leibler divergence of the marginal posterior from the marginal prior, estimated from kernel densities of the samples and expressed in nats. The divergence is zero when the observations say nothing about a parameter; for scale, a posterior with the centre of its prior and a quarter of its width gives about 0.9 nats, and a posterior with the width of its prior moved by one prior standard deviation gives 0.5 nats.

2.13 The comparison with a soil started in balance

To test whether the transient initialisation is needed, every model was calibrated a second time with the soil started in balance, identical in data, error model, priors and sampling, and different in one factor only: in 1985 the soil is set to the state at which its gains and losses balance under the litter input of 1985–1989, scaled by σ_{input}, and the reference climate of 1961–1984. The initial input ratio σ_{init} is fixed at one, which makes the litter path of Section 2.7 flat, and is removed from the parameters; the prior of σ_{input} is unchanged. The balanced start is the steady-state assumption applied at the start of the measurements, and not the procedure of any particular inventory, and is the strongest fair version of that assumption: every other parameter is recalibrated, so the models are free to compensate for the missing deficit through the litter input, the transfer fractions and the response to climate. A soil started in balance is not expected to stay flat, because the litter input rose by about half until 2006; what distinguishes the two starts is the size of the accumulation they produce. The expectations for the comparison were written down before the calibrations were run (Supplementary Section S2.2). The two calibrations are compared on the trajectory of the national mean against the three campaigns, on the rates of change over 1985–2006, 2006–2024 and 1985–2024, on the position of each model along the trade-off between litter input and mean transit time, and on the fit. The comparison rests mainly on the rates rather than on the log-likelihoods, which are directly comparable here but inherit the overconfidence that the error model only partly removes.

2.14 The partitioning of the modelled sink

The sink each calibrated model produces over 1985–2024 was divided among three sources: the state inherited from before 1985, the change of the litter input since 1985, and the climate. One further calibration was made for every model, with the soil started in balance *and* the climate held at its 1961–1984 reference in every year after 1985, in the calibration as in the simulation. Each model is thereby calibrated three times on the same measurements, adding one assumption at a time:

1. the transient initialisation, with the observed litter and climate;
2. a balanced start, with the observed litter and climate (Section 2.13);
3. a balanced start, with the observed litter and the climate held at the reference.

With S_1, S_2 and S_3 the sinks of the three calibrations, the division is

$$S_1 = \underbrace{(S_1 - S_2)}_{\text{inherited state}} + \underbrace{(S_2 - S_3)}_{\text{climate}} + \underbrace{S_3}_{\text{litter change}} ,$$

summarised in Table 2; the three parts add up to the sink of the first calibration by construction.

Table 2: The three parts of the modelled sink. Each part is the difference between two calibrations fitted to the same measurements, which differ in one assumption.

part	calibrations	the assumption that differs	what the part measures
inherited state	$1 - 2$	starting state: estimated, or balanced in 1985	what the deficit inherited from before 1985 adds to the sink
climate	$2 - 3$	climate after 1985: observed, or held at the 1961–1984 reference	what the observed climate adds, acting on the litter that actually arrived
litter change	3	—	the sink of a soil started in balance under an unchanging climate, which exists only because the litter departs from its 1985–1989 level

Each part therefore compares two fitted models that differ by one assumption, including the change of the other parameters that the data demand once the assumption is made, and is not the effect of one source on a model held fixed. The joint effects of climate with the inherited deficit and with the litter change are assigned by the order of the calibrations, to the inherited and to the climate part respectively. The parts are computed for the plots measured in all three campaigns (310 plots) and 50 posterior draws of each calibration, and their additivity and a zero sink under constant litter and climate were checked for every draw. The size of the climate part depends on the reference climate, whereas the ordering of the parts does not. [Five terms (two further calibrations) planned with the new input data.]

2.15 Simulations beyond the observations

The calibrated models were then run beyond the observations. The simulations describe the calibrated models on the study plots, per hectare, under stated forcing, and are not a forecast of the national soil carbon balance. Each model is run for 60 years after 2024 with the litter input held at its 2024 value and the climate of the last 20 observed years repeated three times, so that the projection contains no change of climate or of forest growth; rates over the projection are computed over whole 20-year cycles, because the repeated climate alternates between a cooler and a warmer decade. The projection uses 100 posterior draws and the plots measured in all three campaigns, for both the transient and the balanced-start calibration. The projection is the basis of three further quantities:

2.15.0.1 The carbon remaining to be accumulated before balance.

The equilibrium stock of each plot is the stock at which gains and losses would balance under the litter and climate of the projection, computed as for the mean transit time (Section 2.10); the remaining accumulation is the equilibrium stock divided by the stock of 2024, minus one. The

remaining accumulation is a quantity of carbon, not a time: how long the soil would take to reach balance is set by its slowest pool and is not computed here.

2.15.0.2 What could happen with more litter input.

The projection is repeated with the litter input increased by 20 per cent from 2025. Because the models are linear in their input, the sink over any period is a straight line in the input multiplier, passing through the projections with the input held and increased by 20 per cent; the increase of input that would hold the observed rates of 1985–2024 and of 2006–2024 over the first 10 and 20 years of the projection is read off that line, without further simulation. Without further warming, these increases are lower bounds.

2.15.0.3 What could happen with warming.

A linear rise of temperature, from zero in 2025 to its full value in 2084, is added to the repeated climate, and the stock in 2084 is compared with that of the projection without the rise. The rises are the warming still to come under SSP1-2.6, SSP2-4.5 and SSP5-8.5 for Finland (Ruosteenoja and Jylhä, 2021): the published warming relative to 1981–2010, minus the 1.0 °C by which the repeated climate of 2005–2024 already exceeds that of 1985–2010 on the study plots, giving 1.3, 2.6 and 5.4 °C. The litter input is held constant at its 2024 value, with no response of forest growth, so the result isolates the response of decomposition to warming, and it is a sensitivity, not a projection. The response is computed for three parameterisations of each version of Yasso, each started from its own state under the pre-run climate: the transient and the balanced-start calibrations (40 posterior draws each), and the published parameterisation, from draws of the published posterior (40 posterior draws each) for Yasso15 and Yasso20 and from the published values for Yasso07. None of the published kinetic parameters is refitted. The published posterior samples carry no input parameters for our plots, so both are supplied: the initial input ratio σ_{init} is taken as the posterior median of the transient calibration, and the input multiplier σ_{input} is chosen so that the published model fits the three observed campaign means as closely as possible, fitted for the draw closest to the published median transit time and, for the other draws, scaled by the ratio of their transit times to that of the fitted draw. Because the stock is proportional to the input multiplier, this choice sets the level of the published arm without changing its kinetics, and the response to warming, computed within each draw, does not depend on it.

2.16 Predictive evaluation and residual analysis

The fit of each model to the observations is evaluated with the transient calibration: for 100 posterior draws, the stock is simulated at each date and on each plot where soil carbon was measured, both on the calibration plots and on the held-out plots, which did not enter the calibration. Skill is reported, separately for the two sets, as the coefficient of determination between the observed stocks and the mean of the simulated ones, and as the root mean squared error between the observed stocks and the median of the simulated ones. Residuals are the logarithm of the ratio of the observed to the predicted stock, the scale of the error model; a residual of 0.1 corresponds to about 10 per cent more carbon in the soil than predicted. The residuals of any two models correlate at 0.99 or more, and the detailed analyses use one value per observation, the mean residual of Yasso07 and Yasso15.

These residuals were analysed in three ways. First, a random forest was fitted to the residuals

of each model and evaluated out of bag, with about sixty site variables (climate, soil texture and chemistry, litter composition, the stand in 1985), to measure how much of the residual can be predicted and which variables lead; the ranking was checked with several hundred further variables from the four inventories, soil chemistry, topography, nitrogen deposition, management records and the understorey cover of 2023, leaving out variables that measure soil carbon itself.

Second, the mean residual was regressed on the leading predictor of the random forest, the basal area of the stand in 1985, together with the forest site type, the texture of the mineral soil recorded in 1985 and the volume share of stones.

Third, the residual of each observation was related to the time since the last regeneration cut (a felling that removes the stand) before sampling, dated from the inventories of 1985, 1990, 1995 and 2021–2023 and from the forest use declarations filed with the Finnish Forest Centre from 1997, or, where no cut was recorded, the age of the stand in 1985 plus the years since; the comparison with the basal area of 1985 used a regression with a separate intercept for each plot.

3 Results and discussion

3.1 R1 — The problem, tested: a soil started in balance

3.1.1 What the soil did

3.1.1.1 OPENING Gaining, then slowing

3.1.1.1.1 The rates

3.1.1.1.1.1 Numbers

[+0.40 tC ha^{-1} yr^{-1} over 1985–2006, +0.12 over 2006–2024, +0.26 over the whole span (310 plots measured in all three campaigns, unweighted, at the true sampling years, so that the first interval is about 17 years, not 21).]

REFS: Kramarenko (2012)

3.1.1.1.2 The robust and the fragile interval

[The rate interval of 2006–2024 includes zero; the span is robust. The models are judged mainly on the span and on the first interval.]

3.1.1.2 CHALLENGE A demanding trend for a soil assumed in balance

[A gain that continues while the litter has stopped rising is what a soil in balance with its litter cannot produce: the test the trend sets.]

3.1.2 A balanced start cannot satisfy the three

3.1.2.1 ACTION The test

[The same models, data, priors and error model, with the soil started in balance in 1985 under the 1985–1989 litter and the 1961–1984 climate, and every other parameter recalibrated: the models are free to compensate through the input multiplier, the transfer fractions and the climate response. The strongest fair version of the assumption.]

REFS: Wutzler and Reichstein (2007)

3.1.2.2 ACTION Started in balance, all six models lose carbon

3.1.2.2.1 Over the whole span

3.1.2.2.1.1 Numbers

[1985–2024: −0.11 to −0.02 tC ha^{-1} yr^{-1} against the observed +0.26, in all six models. The models rise with the litter to 2006 and give the rise back after it.]

▶ **Figure 5 here** (`F16_eqinit_trajectories_six`)

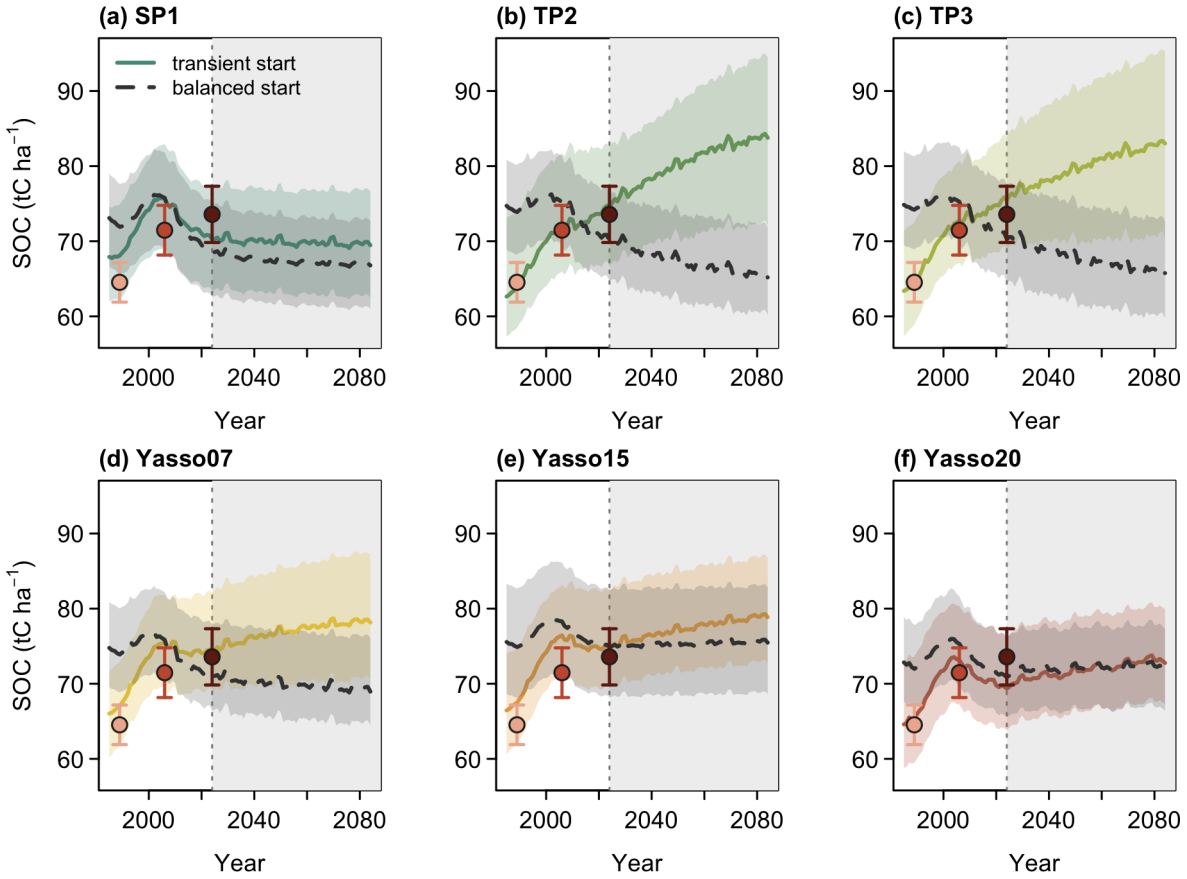


Figure 5: Modelled national mean soil carbon under the two starting assumptions, model by model, against the three campaigns. Coloured line and band: the transient start, with the starting state estimated and dated back to 1917; dashed black line and grey band: the soil started in balance in 1985. Lines are posterior medians and bands 90% intervals of the national mean over posterior draws. Points are the measured campaign means with their 95% confidence intervals, on the 310 plots measured in all three campaigns, unweighted, at the mean sampling year of each campaign. The shaded area after 2024 is the projection, with the litter held at its 2024 value and the climate of 2005–2024 repeated. The two starts differ most at the first campaign, where the balanced start lies above the measurement in every model, and separate again in the projection.

3.1.2.2.2 Above the first campaign from the start

3.1.2.2.2.1 Numbers

[About 9 tC ha^{-1} above the first campaign (73–76 against 64.5): a soil in balance with the 1985 litter holds more carbon than the soil held.]

3.1.2.2.3 Short of the early rise, losing after 2006

3.1.2.2.3.1 Numbers

[1985–2006: +0.06 to +0.20 against +0.40, none within the observed interval; 2006–2024: all six lose carbon. ▷ Supplement, Figure S4 (`F17_eqinit_rates`)]

3.1.2.3 RESOLUTION The trilemma, visualised

3.1.2.3.1 The balanced start gives up the soil series

[Held to the measured litter and to kinetics near the published ones, the balanced start gives up the corner the Introduction predicted: the series.]

3.1.2.3.2 It bends the kinetics, and still fails

3.1.2.3.2.1 How the parameters move

[Less input than the transient start, faster and more temperature-sensitive decomposition (transit time 22–27 → 17–24 yr), and still no accumulation. The expectation written before the runs was the opposite for the input (more, not less): report it.]

3.1.2.3.3 The third corner, partly open

[How far the kinetics would have to move to fit the series with the measured input is left open: a balanced start with widened rate priors (one factor, six jobs), or a limitation?]

3.1.2.4 RESOLUTION The two starts are judged on the rates, not on the fit

3.1.2.4.1 Why the fit cannot decide

3.1.2.4.1.1 Numbers

[The likelihood is dominated by the stock levels, which both starts reach (log-likelihoods within 8, the estimated start better in five of six); the rates and the early stock discriminate. Two sentences, before a reader concludes the starts are equivalent.]

3.1.3 A first round of conclusions

3.1.3.1 RESOLUTION Three things settled

3.1.3.1.1 The trilemma is real on these data

[With the measured litter and near-published kinetics, a soil assumed in balance in 1985 cannot reproduce the measured accumulation.]

3.1.3.1.2 The failure is not a matter of tuning

[The failure survives the freedom to compensate through every other parameter.]

3.1.3.1.3 What is missing is a state out of balance on the start date

[The balanced start lacks a deficit on the start date, a soil still catching up with its inputs.]

3.1.3.2 RESOLUTION The justification of the central choice

[Relax the steady-state assumption and estimate the starting state from the measurements: R2.]

3.1.4 Nested: the uncertainty under a balanced start

3.1.4.1 ACTION The input multiplier relaxes

3.1.4.1.1 Numbers

[Yasso15 1.54 → 1.31, TP2 1.75 → 1.60; SP1 about unchanged. The input multiplier is the point where the uncertainty of the input estimate enters the calibration; under a balanced start the kinetics strain while the input relaxes.]

3.1.4.2 RESOLUTION The initial input ratio is not absent but fixed

[$\sigma_{\text{init}} = 1$ by the steady-state assumption: a value assumed, never tested.]

3.1.4.3 RESOLUTION Mirrored at the end of R2

[One sentence: the same two parameters, now both estimated.]

3.2 R2 — The resolution: relaxing the steady-state assumption

3.2.1 The central choice

3.2.1.1 OPENING From the idea to its implementation

3.2.1.1.1 The idea

[The soil may enter the observations out of balance.]

3.2.1.1.2 The implementation

[The start is estimated from the measurements and dated back to 1917, through σ_{init}, the ratio of the 1917 to the 1985 litter input, the parameter the balanced start fixed at one. Same data, priors and likelihood; one parameter more (Section 2.7).]

3.2.2 The estimated start reconciles the soil series

3.2.2.1 ACTION All six reach the three campaign levels

3.2.2.1.1 Over the whole span

3.2.2.1.1.1 Numbers

[1985–2024: +0.05 to +0.30 against +0.26, four of six inside its interval (SP1 +0.05 and Yasso20 +0.10 below). Figure 5 (coloured lines); the six models together: ▷Supplement, Figure S5 (`F3_mean_soc`)]

3.2.2.1.2 By interval

3.2.2.1.2.1 Numbers

[1985–2006: all six reach the observed rise. 2006–2024: only TP2 and TP3 reach +0.12; Yasso07 near zero (−0.03); SP1, Yasso15 and Yasso20 lose carbon.]

3.2.2.2 RESOLUTION Bracketing is informative because it is not automatic

[Matching the rate would mean little if any calibration on the same data could do it. The balanced start, calibrated on the same measurements with one parameter fewer, fails in every model.]

3.2.3 Where the models land: transit times against the published ones

3.2.3.1 ACTION Calibrated against published

3.2.3.1.1 Faster in Yasso07 and Yasso15, unchanged in Yasso20

3.2.3.1.1.1 Numbers

[Median transit time at the reference: Yasso07 23.7 against a published 33.5 yr (0.71×), Yasso15 26.8 against 30.5 (0.88×), Yasso20 22.3 against 22.1 (1.01×).]

► **Figure 6 here** (`F12_mrt_yasso`)

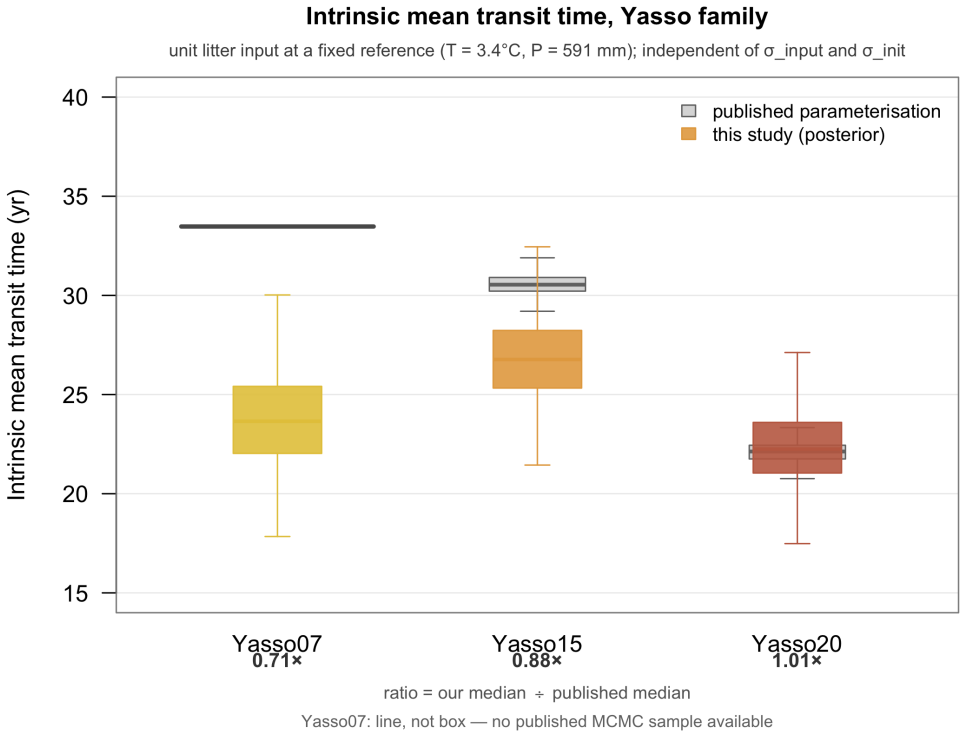


Figure 6: Mean transit time of carbon in the three versions of Yasso, calibrated and published. Transit times are computed from the model equations alone, with a unit litter input at a common reference climate (mean annual temperature 3.4 °C, precipitation 591 mm) and litter composition, so that the comparison depends neither on the input multiplier nor on the starting state. Coloured boxes: the posterior of this study; grey boxes: the published posterior samples of Yasso15 and Yasso20; Yasso07, for which no published sample exists, is drawn as a line at its published value. The ratios below the axis are our median over the published one.

3.2.3.2 RESOLUTION Three different findings, not one

3.2.3.2.1 Yasso07: its single climate modifier

[Yasso07 applies one climate modifier to every pool, so its transit time is the reference value divided by that modifier: the whole gap is the climate response, which the calibration moves.]

3.2.3.2.2 Yasso15 and Yasso20: pool-specific modifiers

[The slow pool keeps its own climate response, which moves little; the gap is small (Yasso15) or absent (Yasso20).]

3.2.4 Across the ridge between input and turnover

3.2.4.1 ACTION The two starts land at different places on the same trade-off

3.2.4.1.1 The soil fixes the product of input and transit time

[Within each start the posterior draws lie along a ridge: more input with faster turnover fits as well as less input with slower turnover.]

3.2.4.1.2 The balanced start sits at less input and faster turnover

3.2.4.1.2.1 Numbers

[Five of six models (SP1, a single slow pool, barely moves); e.g. TP2 from about 4.4 to 4.0 tC ha^{-1} yr^{-1} and from 22 to 17 yr. The likelihood surface: ▷ Supplement, Figure S6 (`F14_mrt_ridge`)]

▶ **Figure 7 here** (`F18_eqinit_tradeoff`)

3.2.4.2 RESOLUTION The trilemma, in parameter space

[The soil series is the ridge; the measured input (the dotted line) and the published kinetics sit off it; a balanced start stands on two of the three at most, the estimated start elsewhere on the same ridge. One or two sentences, without redrawing the figure.]

3.2.4.3 RESOLUTION What the resolution leaves

[The tension concentrates in the input corner: the soil series is met and the kinetics stay near the published ones, while the input the models need stays above what the product supplies (3.2–4.4 against 2.5). Leads straight into the next subsection.]

3.2.5 Unexpected: the input corner gives way

3.2.5.1 CHALLENGE The corner expected to hold

[The Introduction treated the litter input as the most defensible of the three corners. Having crossed the ridge, the models still ask for more input than the trees supply.]

3.2.5.2 ACTION More input than the trees supply

3.2.5.2.1 The size of the gap

[Effective input 3.22–4.40 against 2.511 tC ha^{-1} yr^{-1} of tree litter, i.e. 1.28–1.75×, in all six models.]

3.2.5.2.2 Set by the soil data, not by the prior

[The posterior stays above the prior centre whichever anchor is used: with the higher anchor (1.27) it lands 1.28–1.50× above it (earlier setup; state as such or rerun). The likelihood, not the prior, places the input. Pointer to Supplement S2.1.]

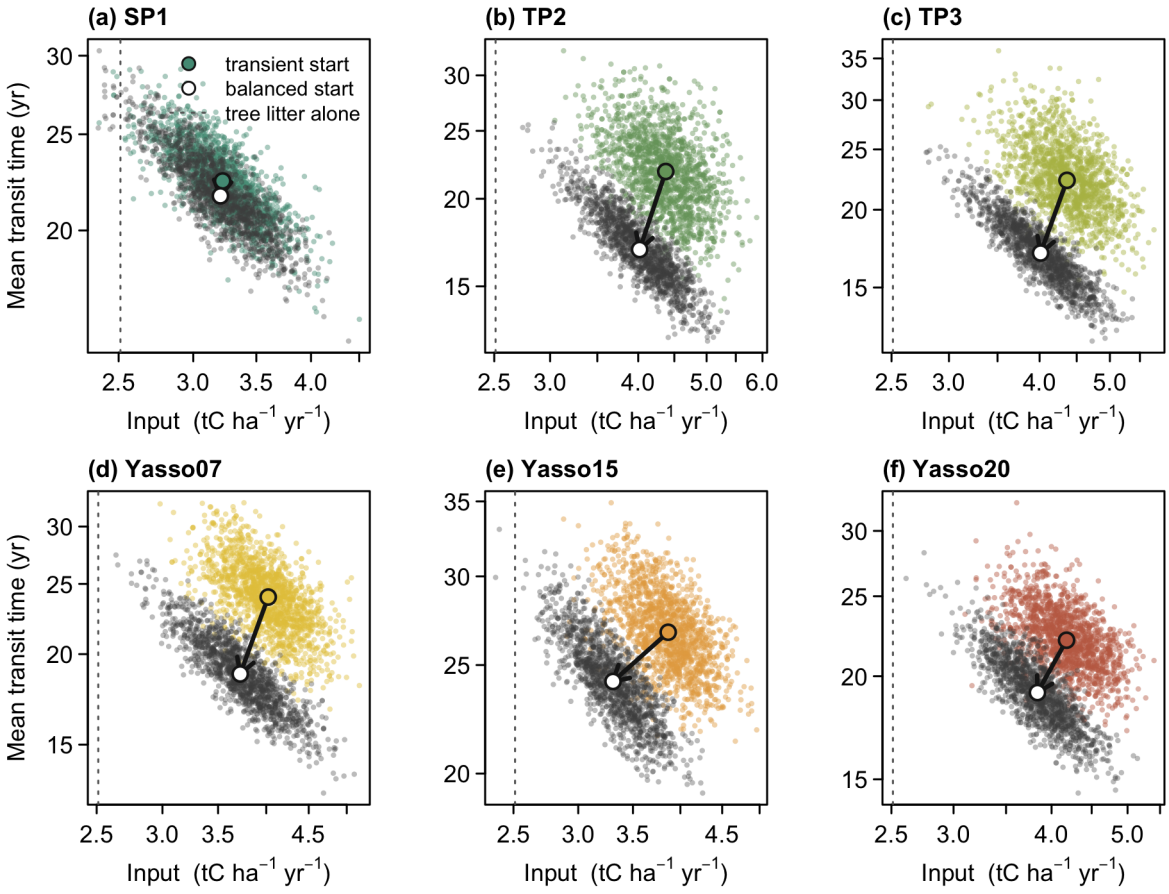


Figure 7: Effective litter input against mean transit time under the two starting assumptions. Points are thinned posterior draws of each model under the transient start (colour) and the balanced start (grey); large points are the medians, joined by an arrow from the transient to the balanced start. The effective input is the input multiplier times the mean litter input of the product; the dotted line marks the tree litter of the product alone (2.51 tC ha^{-1} yr^{-1}). Transit times as in Figure 6. Within each start the draws lie along a ridge of near-constant product of input and transit time; in five of six models the balanced start moves toward less input and faster turnover.

3.2.5.2.3 Not the ridge being paid on the input side

[Yasso20 keeps its published transit time and still needs more input (check its effective input on the current run). The excess is therefore not only the trade-off: it remains where the kinetics are not displaced.]

3.2.5.2.4 Smaller, but present, without the history

[Under the balanced start the input multiplier relaxes (Yasso15 1.54 → 1.31, TP2 1.75 → 1.60, SP1 about unchanged; R1 nested), because a balanced soil already holds its full stock. Part of the gap is the start, part survives it.]

3.2.5.3 ACTION Against independent estimates

[The national inventory's full estimate is 1.08–1.26× the product on the southern plots, bracketing the understorey-corrected anchor (1.21); the plot-level understorey litter gives 1.13. All below the posteriors (1.28–1.75). ▷Supplement, Figure S8 (`SX_input_vs_literature`)]

REFS: Lehtonen and Heikkinen (2016); Gower et al. (2001); Statistics Finland (2025)

3.2.5.4 RESOLUTION A national gap, carried to R4

[The gap is a national level, not a plot pattern: one multiplier, the same in every plot. What could close it (understorey, natural mortality, below-ground inputs) is taken up as hypotheses in Section 3.4; the plot-scale residual there is a separate matter.]

3.2.6 Nested: the uncertainty under the estimated start

3.2.6.1 CHALLENGE The need for the initial input ratio was there before

[Hidden in R1 by the steady-state assumption, not absent. R2 may look more uncertain; it only makes explicit a quantity R1 assumed. ▷Supplement, Figure S7 (`F9_auxiliary_sigmas`)]

3.2.6.2 ACTION The two input parameters

3.2.6.2.1 The input multiplier: more input than measured

3.2.6.2.1.1 Numbers

[Effective flux 3.22–4.40 against 2.511 tC ha^{-1} yr^{-1}, set by the likelihood rather than the prior (the posterior stays above the prior centre whichever anchor is used).] [The finding is in Section 3.2.5; here only the posterior, beside σ_{init}.]

3.2.6.2.2 The initial input ratio: the deficit the data demand

3.2.6.2.2.1 Numbers

[0.45–0.56 in five of six models (SP1 0.78), below the growing-stock record (∼ 0.90).]

3.2.6.2.2.2 How to read it

[An effective starting deficit of the soil (slash-and-burn, raking, heavy cutting) rather than a ratio of litter inputs. It carries the forecast: the remaining accumulation correlates with it at −0.76 to −0.97 (R3).]

REFS: Liski et al. (2006); Peltoniemi et al. (2004)

3.2.6.3 RESOLUTION The error model and the Bayesian points

3.2.6.3.1 Correlated errors, honest information

[Region, plot and campaign offsets integrated out, a declared total of 0.800: the 1205 observations carry about as much information on the national mean as 220 independent ones (check: 220 in the Methods, 221 in the storyline).]

REFS: Oberpriller et al. (2021)

3.2.6.3.2 Priors as information

[Homogenised across models (Table 1); the posterior against the prior shows what the data added.]

3.2.6.3.3 The direction depends on the comparator

[Against the published parameters the calibration turns over faster (Yasso07, Yasso15); against the balanced start, slower. Never add a direction to "displaces along".]

3.2.7 The weight of the assumptions: the division of the sink

3.2.7.1 CHALLENGE How much of the sink does the starting state carry?

[Every part is a comparison between calibrations that differ by one assumption (Table 2), so the division compares assumptions, each answered by the data.]

3.2.7.2 ACTION The sink is mostly inherited

3.2.7.2.1 The six-model mean

3.2.7.2.1.1 Numbers

[1985–2024: of +8.0 tC ha^{-1} gained, inherited state +11.6, litter change +7.3, climate −11.0.]

▶ **Figure 8 here** (`S16_attribution_ensemble_recal`)

3.2.7.2.2 The inherited part exceeds the whole sink in all six

3.2.7.2.2.1 Numbers

[Litter and climate together remove 1.0–5.3 tC ha^{-1}: the warming takes back more than the litter rise adds.]

3.2.7.2.3 Model by model, the picture changes little

[Inherited +6.1 (SP1) to +17.4 (TP2); litter +6.1 to +11.0; climate −8.0 to −15.7. ▷ Supplement, Figure S9 (`S15_attribution_by_model_recal`)]

3.2.7.3 RESOLUTION Necessary, and dominant

3.2.7.3.1 Necessary

[The balanced start shows the history is necessary: with everything else free, the models cannot do without it.]

3.2.7.3.2 Dominant

[The division shows how much it weighs, and answers the objection that the inherited part is large only because the calibration chose a low σ_{init}: the data demand the deficit.]

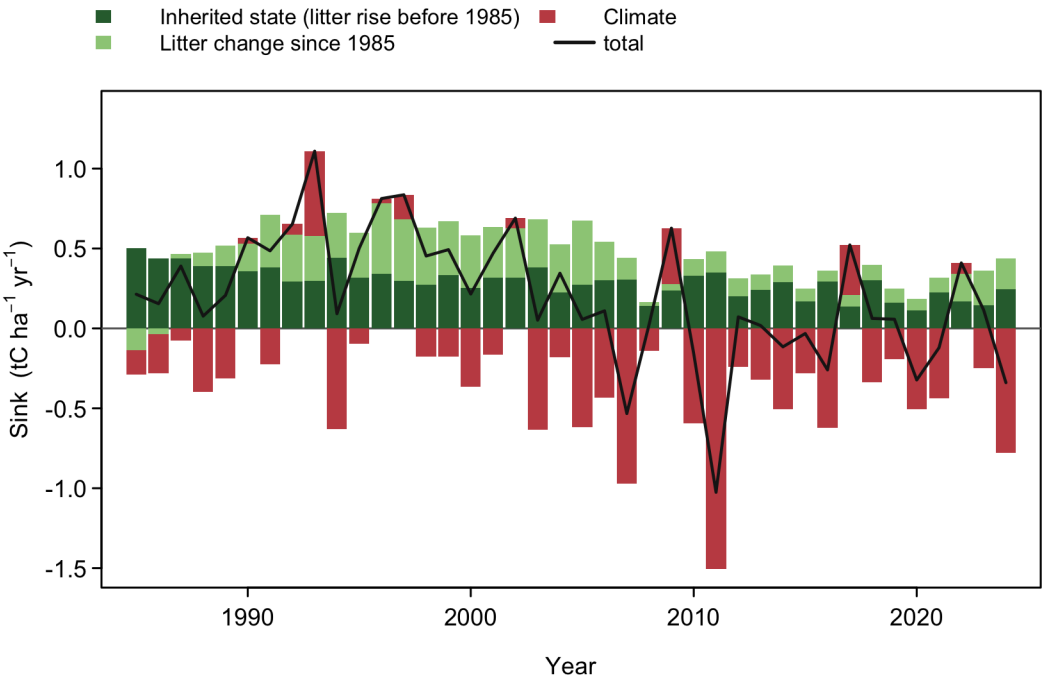


Figure 8: The modelled sink divided into its three parts, year by year, as the average of the six models. Dark green: the inherited state, the part of the sink due to the litter rise before 1985 that the soil had not yet caught up with; light green: the change of the litter input since 1985; red: the climate after 1985; black line: the total sink. Each part is the difference between two calibrations fitted to the same measurements (Table 2). Plots measured in all three campaigns (310), unweighted, 50 posterior draws per calibration. [Three parts until the five-term version exists.]

3.2.7.3.3 The size of the climate part depends on the reference climate

[The ordering is robust (the sink inherited; litter and warming of the same order and opposite sign); the size of the climate part follows the 1961–1984 reference, and a reference that overlaps the observed period would shrink it.]

3.2.7.4 RESOLUTION The bridge to the forecast

[A sink that is the history is a sink that fades: the mechanism of R3. Five terms once the two joint-effect calibrations are run with the new input data.]

3.3 R3 — Consequences for forecasts

3.3.1 The question the resolution raises

3.3.1.1 CHALLENGE Did we get the forecast wrong so far?

[If the starting state decides what the models reproduce, does it decide what they project? Within the scope declared in the objectives: litter held at 2024, recent climate repeated, per hectare on the study plots.]

3.3.2 The starting state decides the forecast

3.3.2.1 ACTION Two starts, two futures

3.3.2.1.1 A balanced soil has no sink left

3.3.2.1.1.1 Numbers

[2025–2044: -0.12 to $+0.02$ tC ha^{-1} yr^{-1}; remaining accumulation to balance -8 to $+9$% of the 2024 stock.]

3.3.2.1.2 A soil carrying its history keeps a sink, which fades

3.3.2.1.2.1 Numbers

[$+0.07$ to $+0.18$; remaining accumulation $+21$ to $+40$% (SP1 ≈ 0). ▷ Supplement, Figure S10 (`F19_eqinit_forecast`) (promote?)]

3.3.2.2 RESOLUTION The disequilibrium found by another route

[Liski et al. (2006) estimated $+38$% from the reconstructed national input. The division of the sink explains the forecast: the sink is the history, and the history fades.]

REFS: Liski et al. (2006)

3.3.3 Side arc: inputs and saturation

3.3.3.1 ACTION Yes, the sink is approaching saturation

3.3.3.1.1 The sink falls while the stock stays below balance

3.3.3.1.1.1 Numbers

[Under the estimated start, with the input held at 2024, the sink falls by a third to a half between the first and the last climate repetition (2025–44 to 2065–84): Yasso07 0.085 → 0.042, Yasso15 0.087 → 0.051, Yasso20 0.074 → 0.048, TP2/TP3 0.18 → 0.10 tC ha^{-1} yr^{-1}; already in 2025–44 the versions of Yasso stay below the observed 2006–2024 rate ($+0.12$). The carbon still to be gained before balance is 25–40% of the

2024 stock in the versions of Yasso (21% in TP2/TP3) and still 19–33% in 2084. SP1 is already at balance (its sink ≈ 0). ▷ Supplement, Figure S11 (`S_saturation`)]

3.3.3.1.2 Saturation of the sink, not of the stock

[The fast pools approach balance first; the remaining distance lies in the slow pool, which closes it over centuries.]

REFS: Sierra et al. (2018)

3.3.3.2 ACTION How much input would postpone it

3.3.3.2.1 About a tenth more, sustained

3.3.3.2.1.1 Numbers

[About 10% more input holds the long-term observed sink (+0.26) for twenty years in the three versions of Yasso, 0.4–0.5 tC ha^{-1} yr^{-1} of effective input; each +10% buys 0.15–0.24 tC ha^{-1} yr^{-1} of sink.]

▶ Table 3 here

3.3.3.2.2 Which target

[Observed long-term rate, recent rate, or each model's own? A warming-corrected column? National units need a sourced area and Aleksi's review.]

3.3.3.3 RESOLUTION Saturation comes regardless

3.3.3.3.1 A higher input raises the balance, and the new sink fades too

3.3.3.3.1.1 Numbers

[With +20% from 2025, the versions of Yasso realise about half of the eventual gain within 60 years (55 / 49 / 46%; TP2/TP3 58%), and the gain lasts only while the input does. The raised input first lifts the sink above the observed long-term rate in all six (0.39–0.43 in 2025–44), then lets it fall back to 0.08–0.12 in the versions of Yasso by 2065–84, i.e. to the observed recent rate. ▷ Supplement, Figure S11 (`S_saturation`) (promote?)]

3.3.3.3.2 Input postpones saturation; it does not remove it

3.3.3.3.3 Lower bounds

[No further warming in these projections: under warming, the input needed is larger. The required input depends on both the starting state (the baseline sink) and the transit time (the gain per unit of input).]

Table 3: The increase of litter input that would hold a given sink, in per cent of the calibrated input, sustained from 2025. Computed from the projections with the input held and increased by 20 per cent, using the linearity of the models in their input (Section 2.15); the projections carry no further warming, so the values are lower bounds. [Numbers from `doublechecks/input_to_hold_sink.R`, runs of 2026-10-12; provisional.]

target sink	period	Yasso07	Yasso15	Yasso20	TP2 / TP3	SP1
observed 1985–2024 (+0.26)	10 yr	+8%	+7%	+5%	+5 / +5%	+11%
	20 yr	+11%	+10%	+11%	+7 / +6%	+13%
each model's own 1985–2024	20 yr	+9%	+7%	+3%	+10 / +10%	+4%
observed 2006–2024 (+0.12)	20 yr	+2%	+2%	+3%	−5 / −5%	+6%

3.3.4 Side arc: the sensitivity to temperature

3.3.4.1 CHALLENGE Does the starting assumption change how the soil responds to warming?

[The balanced start settles at faster, more temperature-sensitive decomposition (R1); the published parameters are a second point of comparison.]

3.3.4.2 ACTION Warming takes carbon in every version of Yasso

3.3.4.2.1 Numbers

[Linear warming to SSP2-4.5 (+2.6 °C still to come by 2084): −5 to −7 tC ha^{-1} by 2084 against the stationary projection. Litter frozen: a sensitivity of decomposition, biased toward loss. SSP1-2.6 and SSP5-8.5 scale the same pattern.]

REFS: Ruosteenoja and Jylhä (2021)

▶ **Figure 9 here** (`F15_forward_arms`)

3.3.4.3 ACTION The starting state moves the trajectory, hardly the sensitivity

3.3.4.3.1 Numbers

[Transient minus balanced start, SSP2-4.5: Yasso07 −0.2 [−2.3, 2.2], Yasso15 −0.9 [−2.1, 0.3], Yasso20 −1.0 [−2.4, 0.4] tC ha^{-1} (90% intervals).]

3.3.4.4 RESOLUTION Only Yasso07 separates from its published parameters

3.3.4.4.1 Its single climate modifier ties turnover to temperature sensitivity

3.3.4.4.1.1 Numbers

[Transient minus published: −2.7 tC ha^{-1} [−4.2, −1.4]; Yasso15 and Yasso20 cross zero.]

REFS: Tuomi et al. (2009); Viskari et al. (2022)

3.3.4.4.2 The applied point

[The inventory runs Yasso07: its response to warming depends on which parameters it runs with, more than on how it is started.]

3.4 R4 — Future development: why the models fail locally

3.4.1 Good trends, poor local skill

3.4.1.1 ACTION The skill numbers

3.4.1.1.1 Numbers

[Calibration R^2 0.015–0.020, held-out $R^2 \leq 0.001$; skill does not climb the pool ladder (SP1 to Yasso20). ▷ Supplement, Figure S12 (`F6_obs_vs_pred_holdout`)]

3.4.1.1.2 Against the spread of the observations

[Optional: the observed stocks vary between plots with a coefficient of variation of 0.36–0.48, which the models do not order.]

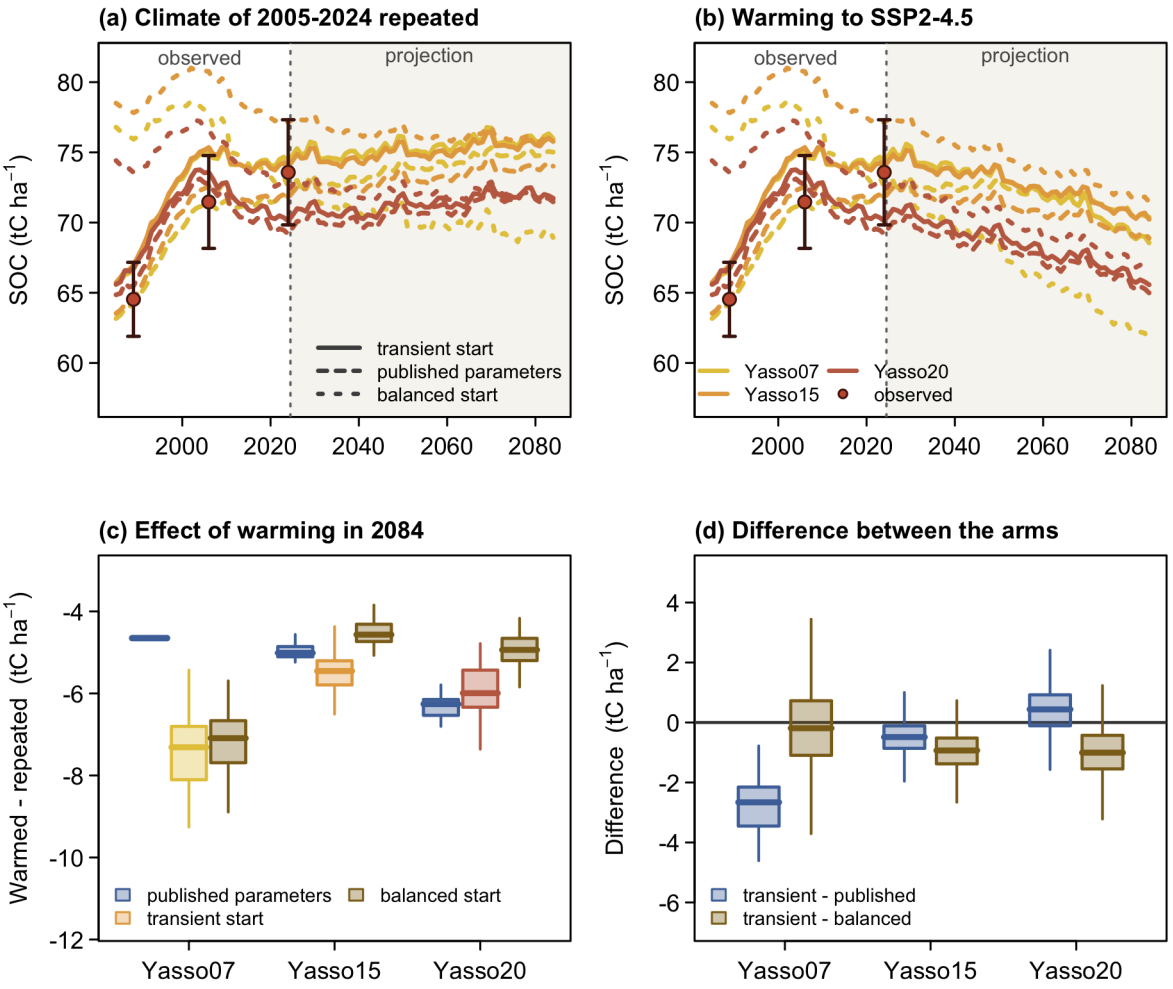


Figure 9: The response of soil carbon to warming under three parameterisations of each version of Yasso. (a) Modelled national mean stock with the climate of 2005–2024 repeated after 2024, and (b) with a linear warming added from 2025 to +2.6 °C in 2084, the warming still to come under SSP2-4.5 (Ruosteenoja and Jylhä, 2021); posterior medians for the transient start (solid), the balanced start (dotted) and the published parameters (dashed), against the measured campaign means. (c) The change of stock in 2084 due to warming, (b) minus (a), computed within each parameter set; boxes span the interquartile range of 40 draws, and the published parameters of Yasso07 are a single set. (d) The difference in that change between the transient calibration and the published parameters (blue) and between the transient and the balanced start (brown); a box that crosses zero means that the choice changes nothing. The litter input is held at 2024, so the response is that of decomposition alone. The input multiplier of the published arm is set to fit the three campaign means, so its agreement with them in (a) is by construction and not a better fit.

3.4.1.2 CHALLENGE What the national fit hides

[The models reproduce the national trend and almost none of the differences between plots: the question of R4.]

3.4.2 The inputs and the soil disagree between plots

3.4.2.1 CHALLENGE The trilemma at plot scale

3.4.2.1.1 Numbers

[Across the range of basal area the litter product varies about ×18, the measured soil about ×1.2; the models pass the litter gradient through (the models do not amplify it). A soil in balance with its litter would follow the litter; the soil does not. One or two sentences.]

3.4.3 What the residual follows: each plot carries its own history

3.4.3.1 ACTION The stand of 1985 leads

3.4.3.1.1 The random forest

[On each model's residual, out-of-bag R^2 0.44–0.65: basal area of 1985 first, then the development class of 1985 and the coarse fragments; several hundred further variables (defoliation, damage, management, nitrogen deposition, topographic wetness, understorey cover) add nothing.]

▶ **Figure 10 here** (`F11_rf_heatmap`)

3.4.3.2 ACTION The stand and the soil

3.4.3.2.1 After basal area

[Basal area alone explains 42% of the residual variance; fertility, texture and stones raise the share to 53%. Fertile sites +14%, xeric −26%; fewer stones +29%, many −17% relative to the average plot.]

▶ **Figure 11 here** (`F11_stand_and_soil`)

3.4.3.2.2 The stone effect

[Check whether the stone effect is partly the stone correction of the observed stocks.]

3.4.3.3 RESOLUTION The excess fades in the same plots

3.4.3.3.1 At plot scale, a wrong starting state rather than a missing input

3.4.3.3.1.1 Numbers

[In the 88 plots with less than 10 $m^2\,ha^{-1}$ in 1985, the residual falls +0.91 → +0.53 → +0.28 over the three campaigns; above 20 $m^2\,ha^{-1}$, −0.44 → −0.41 → −0.29. An error largest at the start and shrinking is the signature of the starting state; a missing input would persist or grow.]

3.4.3.3.2 One national pre-run shape cannot carry each stand's history

[A stand regenerated in the 1960s starts in balance with its low young-stand litter while its soil still holds the previous stand's carbon; mature stands the reverse.]

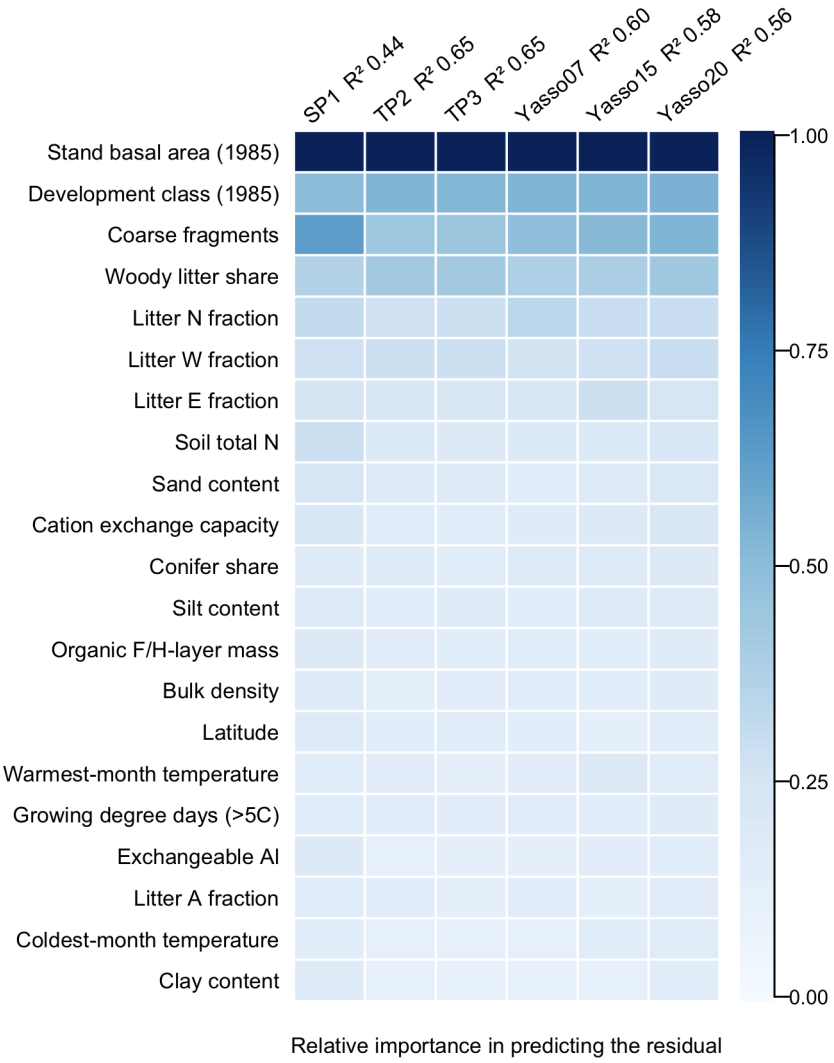


Figure 10: Site variables that predict what the models leave behind. Relative importance of each site variable in a random forest fitted to the residuals of each model, scaled within each model so that the most important variable equals one; the out-of-bag coefficient of determination of each forest is given above its column. The same variables lead in all six models: the basal area of the stand in 1985 first, then the development class of the stand in 1985 and the content of coarse fragments in the soil.

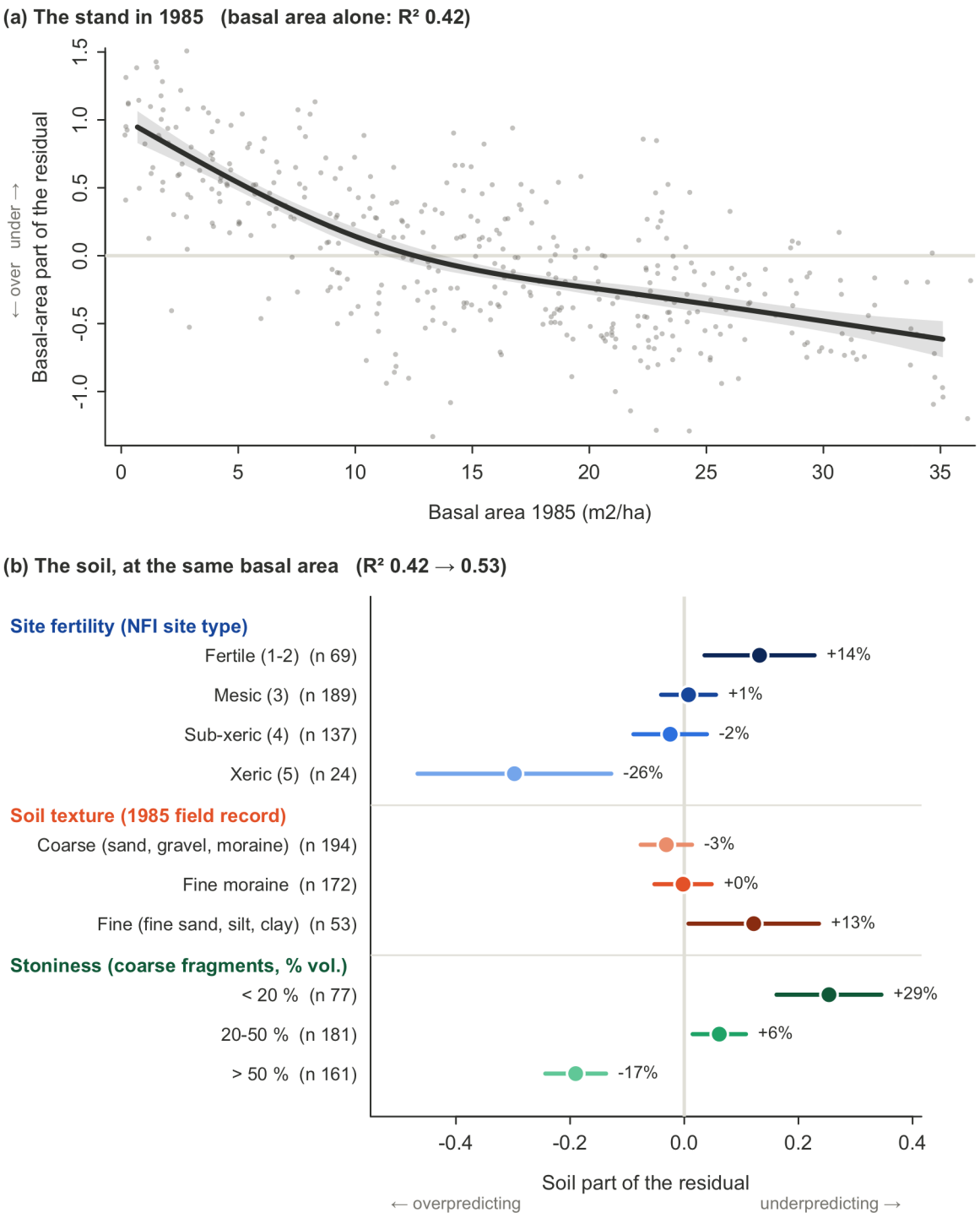


Figure 11: What the models leave behind, in two parts: the stand of 1985 and the soil. The residual is the logarithm of the measured over the predicted soil carbon, as the mean of Yasso07 and Yasso15 over the observations of each plot (419 plots); a positive residual means that the soil holds more carbon than the models predict. One regression on the basal area of the stand in 1985 (a smooth curve) and on the fertility, texture and stone content of the site divides the predicted residual of each plot into parts, each measured from the average plot. (a) The part due to basal area, with its 95% confidence band; the points are the plots. (b) The part due to each property of the site, with 95% confidence intervals, at the same basal area; the labels translate each part into soil carbon relative to the average plot. Basal area alone explains 42 per cent of the residual variance, and the three properties of the site raise the share to 53 per cent.

3.4.4 What the models miss: hypotheses

3.4.4.1 OPENING Hypotheses, with the literature mixed with our results

[Speculative by design; each hypothesis with the evidence we have and the literature that supports or limits it.]

3.4.4.1.1 Two gaps at two scales

[The national input gap (Section 3.2.5) and the plot residual are different findings: the first is a level shared by all plots, the second a pattern that fades with time. The starting state explains the second; the missing fluxes below are candidates for the first.]

3.4.4.2 RESOLUTION A starting state per plot

3.4.4.2.1 The error fades at the models' turnover pace

[The decay of the plot error takes about 30 years, close to the models' transit time.]

3.4.4.2.2 The excess lives in stands young in 1985

3.4.4.2.2.1 Time since the last regeneration cut

[The residual peaks 20–30 years after the cut, and only in stands that were sparse in 1985; dated cuts after 1985 show no peak: the signal belongs to the starting state, not to cutting.]

▶ **Figure 12 here** (`F11b_time_since_cut`)

3.4.4.3 RESOLUTION Understorey

3.4.4.3.1 Numbers

[Field-layer cover of 2023 (438 of 456 plots) adds a little after basal area (graminoid and field-layer cover positive, about $\Delta R^2 = 0.01$); the plot-level understorey litter gives (J+u)/J $\approx$ 1.13, below the posteriors (1.28–1.75).]

REFS: Muukkonen and Mäkipää (2006); Muukkonen et al. (2006); Lehtonen and Heikkinen (2016)

3.4.4.4 RESOLUTION Natural mortality and below-ground inputs

3.4.4.4.1 Natural mortality

[0.10–0.16 tC ha^{-1} yr^{-1} by now in the south in the national inventory, absent from the product.]

REFS: Statistics Finland (2025)

3.4.4.4.2 Roots, rhizodeposition, mycorrhizal fungi, fine-root lifespan

[Exudates and mycorrhizal necromass add input; a longer fine-root lifespan would point the other way (`literature/05-07`).]

REFS: Bolinder et al. (2007); Clemmensen et al. (2013); Helmisaari et al. (2007); Strand et al. (2008)

3.4.4.4.3 A set that brackets the input gap, never a sum that matches it

[Understorey about +13%, natural mortality 0.10–0.16 tC ha^{-1} yr^{-1} (about +4–6%): together about 1.18–1.19×, and the inventory's full estimate at most 1.26×, all short of the lowest posterior (1.28). Roots, exudates and fungi are the only candidates for the remainder, and their sign is uncertain. Each is a candidate; no

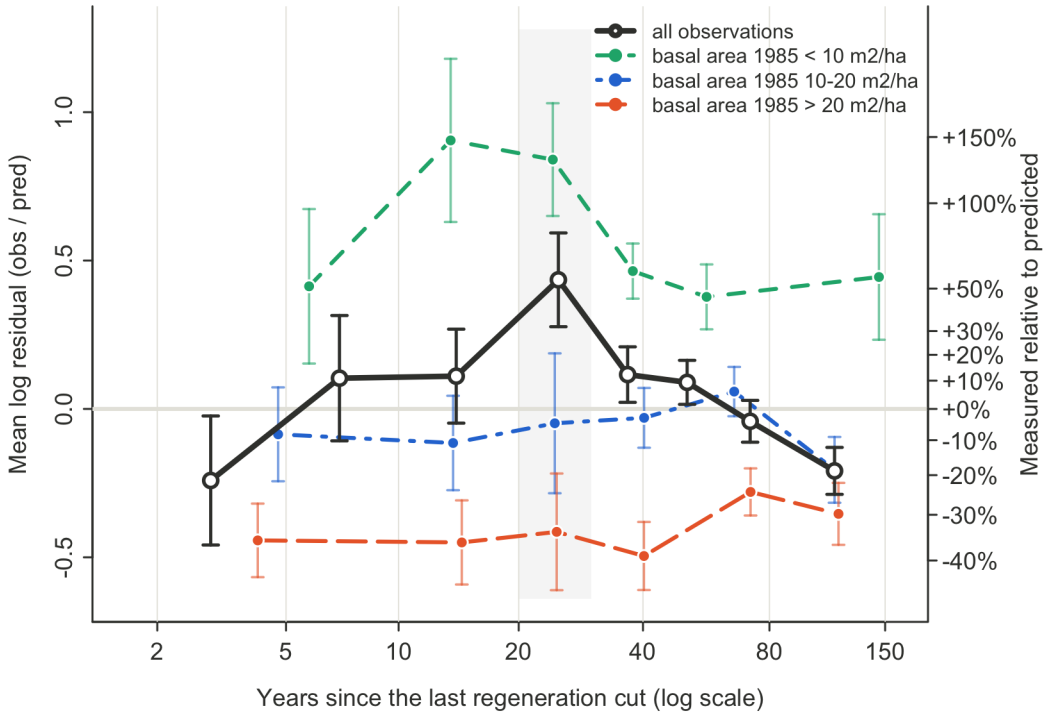


Figure 12: What the models leave behind, against the time since the last regeneration cut. The residual is defined as in Figure 11, here for each soil observation (1172 observations), with binned means and 95% confidence intervals. The time since the cut is taken from the dated cuts in the stand records and the forest use declarations where a cut precedes the sampling, and otherwise as the age of the stand in 1985 plus the years elapsed since. The solid line is all observations; the dashed lines divide them by the basal area of the stand in 1985, with wider bins. The right axis translates the residual into soil carbon relative to the prediction. The shaded band marks 20 to 30 years after the cut.

combination is fitted to the gap. NOTE: the measured fluxes do NOT bracket the gap, so the title must change unless the below-ground terms carry it: decide.]

3.4.4.5 RESOLUTION Pioneer vegetation, where the two meet

3.4.4.5.1 Inherited, or supplied by vegetation no product records

[The excess carbon of stands young in 1985 was built during early stand development, either inherited or supplied by pioneer vegetation no tree-based product records; the data do not separate the two. Cite Lauri Mehtätalo (LUKE): which work?]

REFS: Rana et al. (2023); Mehtätalo et al. (2025); Palviainen et al. (2005)

3.4.4.5.2 What the models do after a cut

[Litter, stock and heterotrophic respiration around a dated cut; modelled respiration against stand age, to set beside flux-tower data (compare trajectories only). ▷Supplement, Figure S13 (`harvest_dynamics`), ▷Supplement, Figure S14 (`rh_vs_stand_age`)]

REFS: Kolari et al. (2004); Kulmala et al. (2011); Peichl et al. (2022)

3.4.5 Suggestions for the models

3.4.5.1 RESOLUTION A short list

3.4.5.1.1 A starting state per plot from each stand's dated history

3.4.5.1.2 Understorey and natural mortality in the litter input

3.4.5.1.3 An early-succession input after regeneration cuts

3.4.5.1.4 The site, where litter alone cannot order the soil

[Fertility and stones.]

4 Conclusions

4.0.0.1 Yes, the forest soil sink is approaching saturation

4.0.0.1.1 The remaining sink is inherited and fades

[By a third to a half within 60 years, although the stock stays 21–40% below balance: the sink saturates, not the stock.]

4.0.0.1.2 More input postpones saturation, but does not remove it

[About a tenth more holds the past sink for twenty years.]

4.0.0.2 The steady-state assumption shows its limits under a complex forestry history

4.0.0.2.1 Started in balance, the models cannot reconcile the three

[The soil series, the kinetics and the inputs.]

4.0.0.2.2 Carrying the history, they can, as far as the data allow

[The tension concentrates in the input corner.]

4.0.0.3 The starting state has consequences for forecasts

4.0.0.3.1 A soil assumed balanced has no room left

[A soil that carries its history keeps a fading sink.]

4.0.0.3.2 The response to warming depends on the parameters more than on the start

[Where turnover and temperature sensitivity are tied, as in Yasso07, the inventory's model.]

4.0.0.4 The models need improving for the spatial variance

4.0.0.4.1 A starting state per plot

4.0.0.4.2 The inputs no tree-based product records

Acknowledgements

[Acknowledgements: LUKE Data Support (climate data); the soil carbon data providers.]

Data and code availability

[Data and code availability: the soil carbon measurements (not a published dataset; settle with the holders), the litter product (Zenodo 10.5281/zenodo.19736499), the climate (FMI via the LUKE weather database), the code repository, the SOC maps (Zenodo).]

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Supplementary material

S1 Data

S1.1 The published Yasso07 against the campaigns

[Where does this figure belong? It shows the published model before calibration, not the soil carbon basis the Methods point it to; R1 is the natural place.]

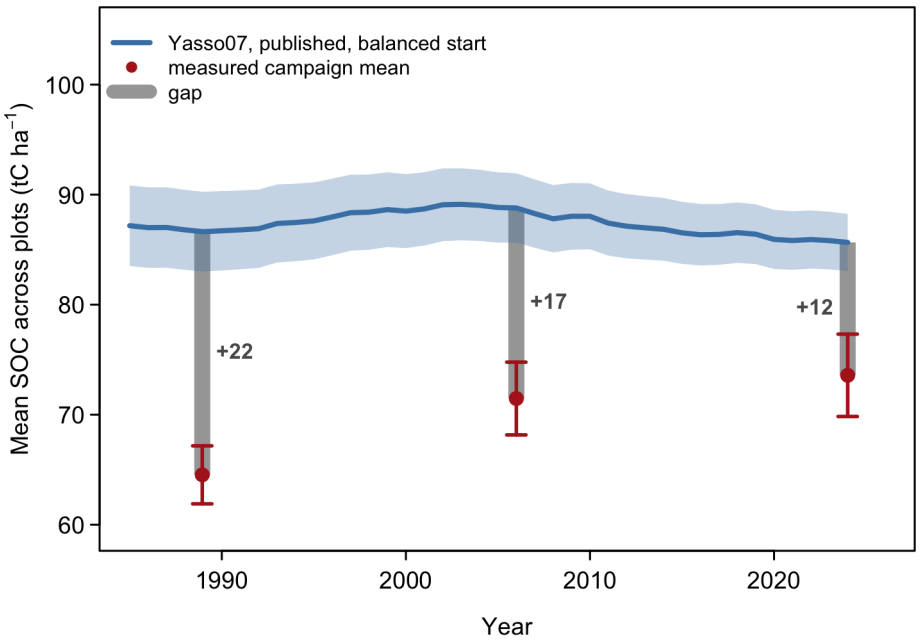


Figure S1: Yasso07, the model of the Finnish inventory, before calibration: the published parameter values, a soil started in balance and the litter of the product, against the three campaign means. Line and band: the mean modelled stock with its standard error, over the 310 plots measured in all three campaigns; points: the measured campaign means with 95% confidence intervals, on the same plots, unweighted. The model is started in balance in 1985 under the 1985–1989 litter and the 1961–1984 climate, with the published parameters of Tuomi et al. (2009, 2011) and no correction of the input. The modelled stock stays nearly flat (87.2, 88.8 and 85.7 tC ha^{-1} at 1985, 2006 and 2024) while the soil gains, so the model starts 22 tC ha^{-1} above the first campaign and the gap closes to 12 only because the soil rises toward it: a soil started in balance has no deficit to recover from, and its level is the litter of the product times the published transit time (33.5 years, Figure 6). [Move to R1 as the first picture of the problem? Same figure for Yasso20 (below the soil, −11 to −21) with `HIKET_F2_MODEL=Yasso20`.]

S1.2 The measured stock in the south and the north

S1.3 The litter product against the national inventory

S1.3.0.1 The shared scope

[Tree turnover and harvest residues; the inventory's natural mortality and ground vegetation excluded.]

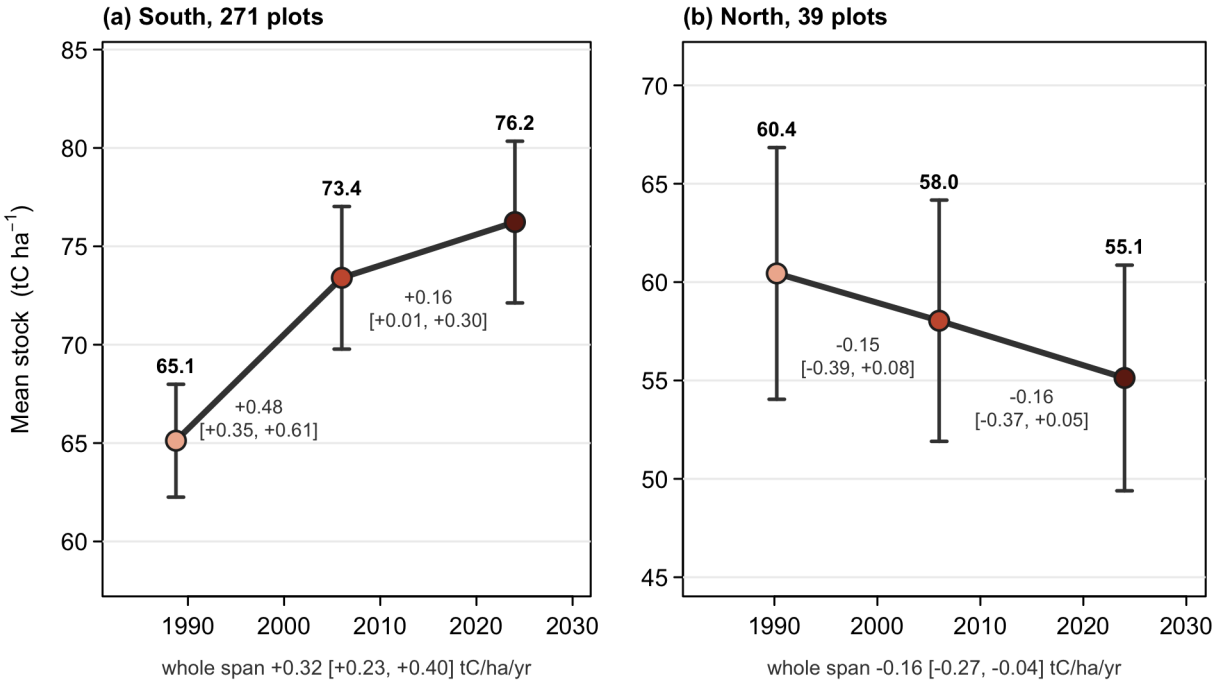


Figure S2: The measured soil carbon stock of Figure 2, separately for the southern and the northern plots. (a) The 271 southern plots; (b) the 39 northern plots, above about 66°N, which the national network samples at a third of the southern density. Points are the mean stock across the plots measured in all three campaigns, with the 95% confidence interval of the mean, and each campaign is drawn at the mean year in which its plots were sampled. Each segment is labelled with the rate of change in tC ha^{-1} yr^{-1} and its 95% confidence interval, from the paired difference between campaigns on the same plots; the rate over the whole period is given below each panel. The two panels share the length of the vertical axis, so the slopes can be compared directly. The stock is the whole soil profile, as in Figure 2. The south gains throughout and slows after 2006 (+0.48, then +0.16); the north loses carbon at an even pace (−0.15, then −0.16; whole period −0.16, 95% interval −0.27 to −0.04).

S1.3.0.2 South and north

[Agreement in the south, where 406 of the 456 plots lie; in the north the inventory lies higher and keeps rising (50 plots).]

S1.3.0.3 Natural mortality

[In the inventory only, and growing.]

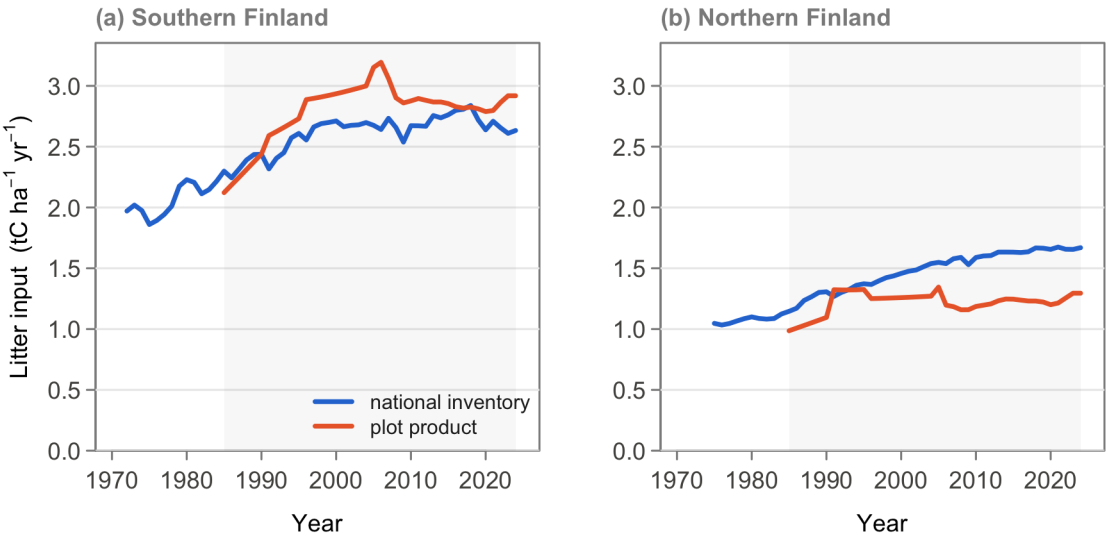


Figure S3: The litter input of the plot product against the national inventory's estimate, on the scope they share. Litter from the turnover of living trees and from harvest residues, per hectare: the national inventory (blue, per hectare of mineral forest land) and the plot product used here (red, the mean of the study plots), in southern (a, 406 plots) and northern Finland (b, 50 plots). The inventory's terms for natural mortality and ground vegetation are excluded. Shaded: 1985–2024.

S2 Methods in detail

S2.1 The prior of the two input parameters

S2.1.0.1 The construction

[A normal prior with standard deviation 0.25 on the logistic scale of the litter flux inside 0.05–8.7 tC ha^{-1} yr^{-1} (Gower et al. 2001), for the present and the 1917 flux.]

S2.1.0.2 The resulting priors

[σ_{input} median 1.07, 95% 0.75–1.45; σ_{init} median 0.93, 95% 0.58–1.49. A figure of the two priors?]

S2.2 The expectations written before the balanced-start calibrations

S2.2.0.1 The four expectations and the refuting one

[From `NEXT_RUN_equilibrium_init.md`, written 2026-10-08: both starts reproduce the campaign levels; the balanced start accumulates too little; it compensates along the trade-off with more input, shorter transit

time or both; Yasso07 has the most room. Refuting: the balanced start reproduces the observed rates at no cost in fit.]

S2.2.0.2 Which held

[Two did not: the balanced start overshoots the first campaign; its input multiplier went down, not up.]

S2.3 The exact yearly solution of the simple models

[SP1 $C_{t+1} = C^* + (C_t - C^*)\, e^{-k\xi}$, $C^* = I/(k\xi)$; TP2 and TP3 as cascades of one exponential per pool.]

S2.4 The rates of ICBM and the climate of Ultuna

[$\xi_{\text{Ultuna}} = 0.94$ at 5.4 °C, amplitude 10 °C, 520 mm; sensitivity to the amplitude convention (9–12 °C).]

S2.5 The division of the sink, in words

[From `appendices/appendix_attribution.tex`: the three calibrations, the meaning of each part, the checks.]

S3 R1: the balanced start

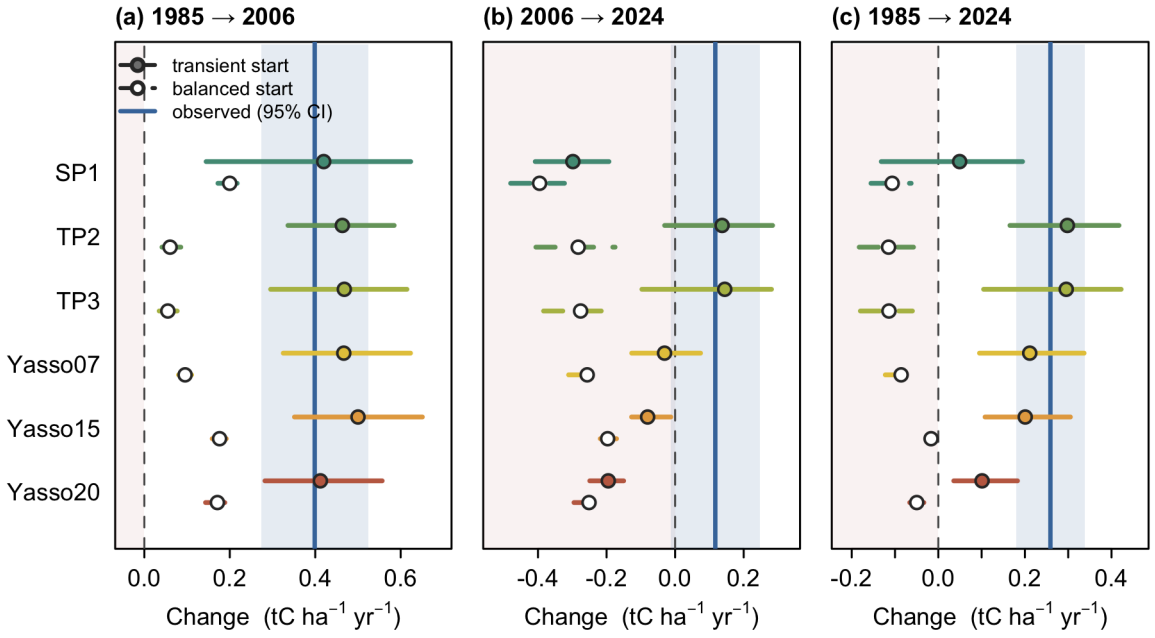


Figure S4: Modelled against observed mean annual change of soil carbon over 1985–2006, 2006–2024 and 1985–2024. Filled points: the transient start; open points: the balanced start; bars: 90% intervals over posterior draws of the national mean. Blue line and band: the observed rate with its 95% confidence interval. Plots measured in all three campaigns (310), unweighted, whole profile.

S4 R2: the estimated start

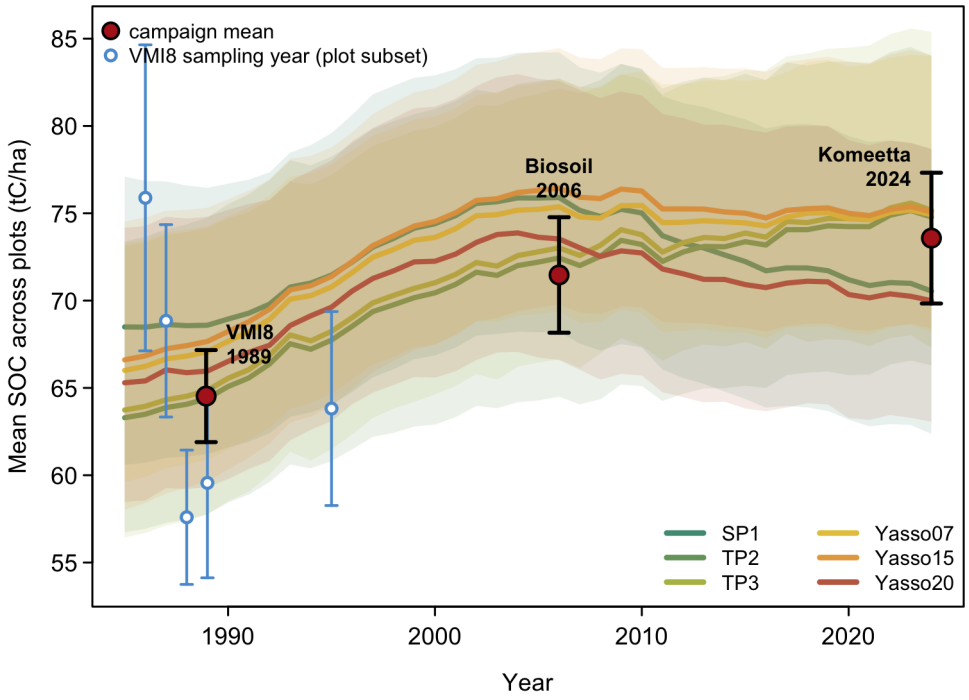


Figure S5: Modelled and measured mean soil carbon, 1985–2024, under the transient start. Lines are the mean stock across plots for each of the six models, and the shaded bands are 95% posterior intervals. Red points are the measured campaign means with their confidence intervals, on the same plots as the model curves; open blue points are the means of the first campaign by sampling year, each a different subset of plots. [Check the plot set of the curves against Figure 5.]

S5 R3: forecasts

S6 R4: the plots

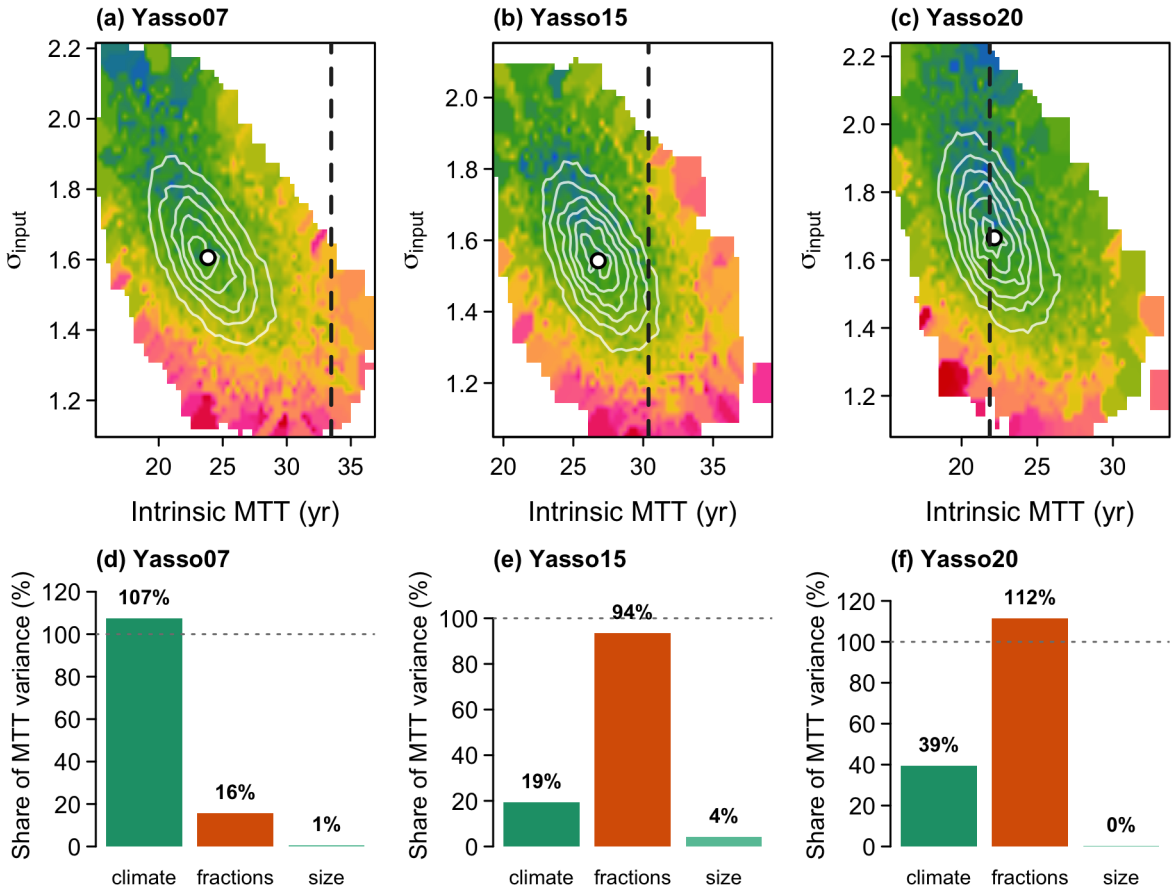


Figure S6: The trade-off between transit time and litter input on the likelihood surface. For each version of Yasso, the best fit to the data attainable at each combination of mean transit time and input multiplier. The colours show the fit to the data alone, with the priors excluded, as the deficit of log-likelihood from the best fit found, from 0 (blue) to 25 units or more (red); the white contour lines show the posterior, which combines the fit with the priors. The dashed line marks the published transit time and the white dot the posterior median. Cells visited fewer than five times are left blank: not visited, rather than impossible to fit. (d)–(f) The share of the posterior variance of the transit time due to each group of parameters (first-order indices; a total above 100% means that the groups interact).

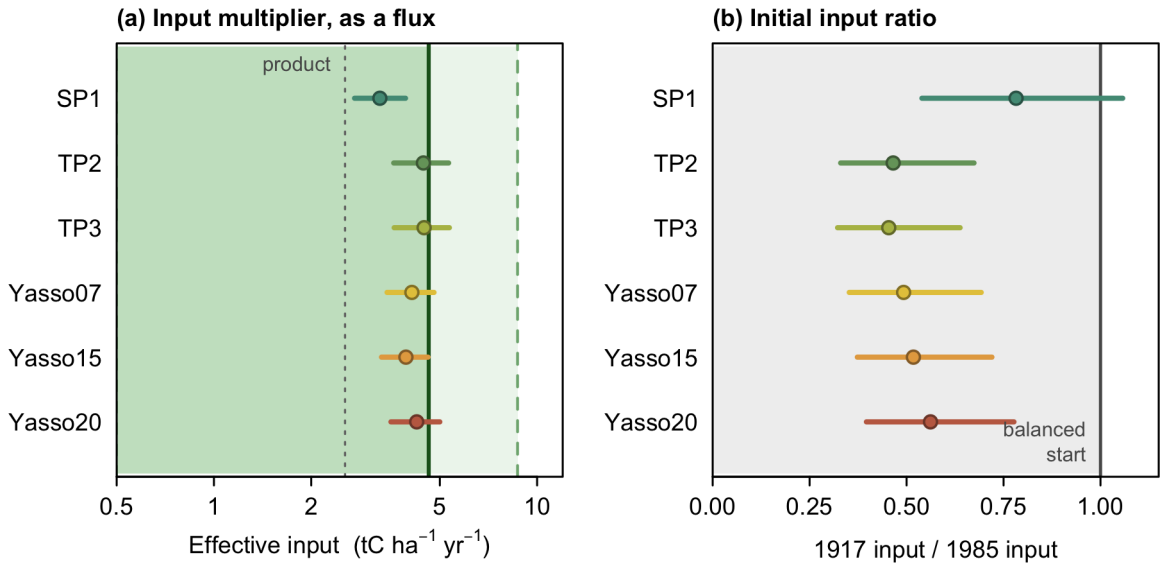


Figure S7: The two input parameters, read as physical quantities. Left: the input multiplier expressed as an effective flux of litter carbon to the soil, the multiplier times the input the product supplies; the dotted line marks the flux of the product itself, and the shaded bands the envelope of independent estimates of boreal net primary production. Right: the initial input ratio, the 1917 litter input over the 1985 input. [Promote?]

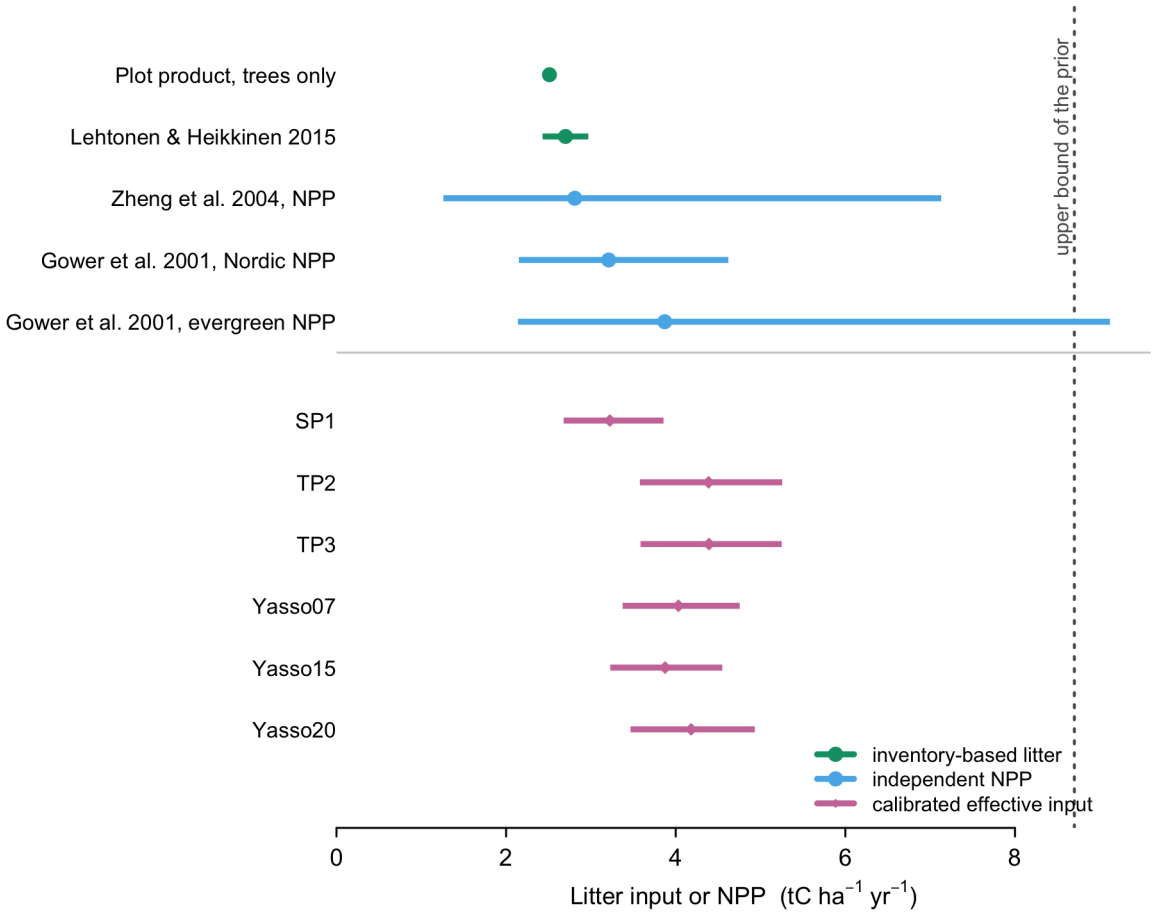


Figure S8: The litter input inferred by the calibration, against independent estimates of the same flux. Green: inventory-based litter products; blue: independent estimates of net primary production, which bound the litter flux from above; lower group: the effective flux inferred here for each of the six models. Bars are reported ranges for the literature and 95% posterior intervals for the models; the dotted line marks the upper bound of the prior on the effective input (Section S2.1).

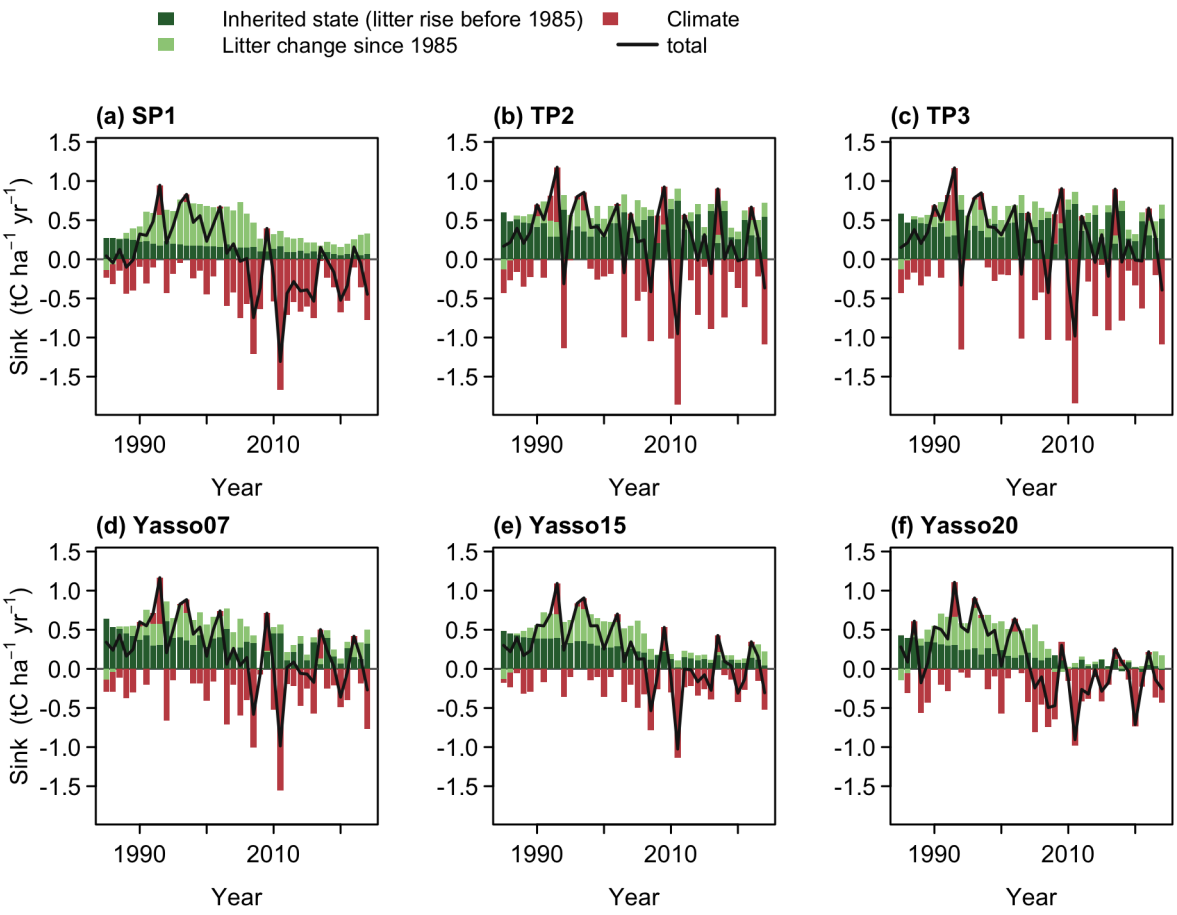


Figure S9: The modelled sink divided into its three parts, model by model. As Figure 8, for each of the six models.

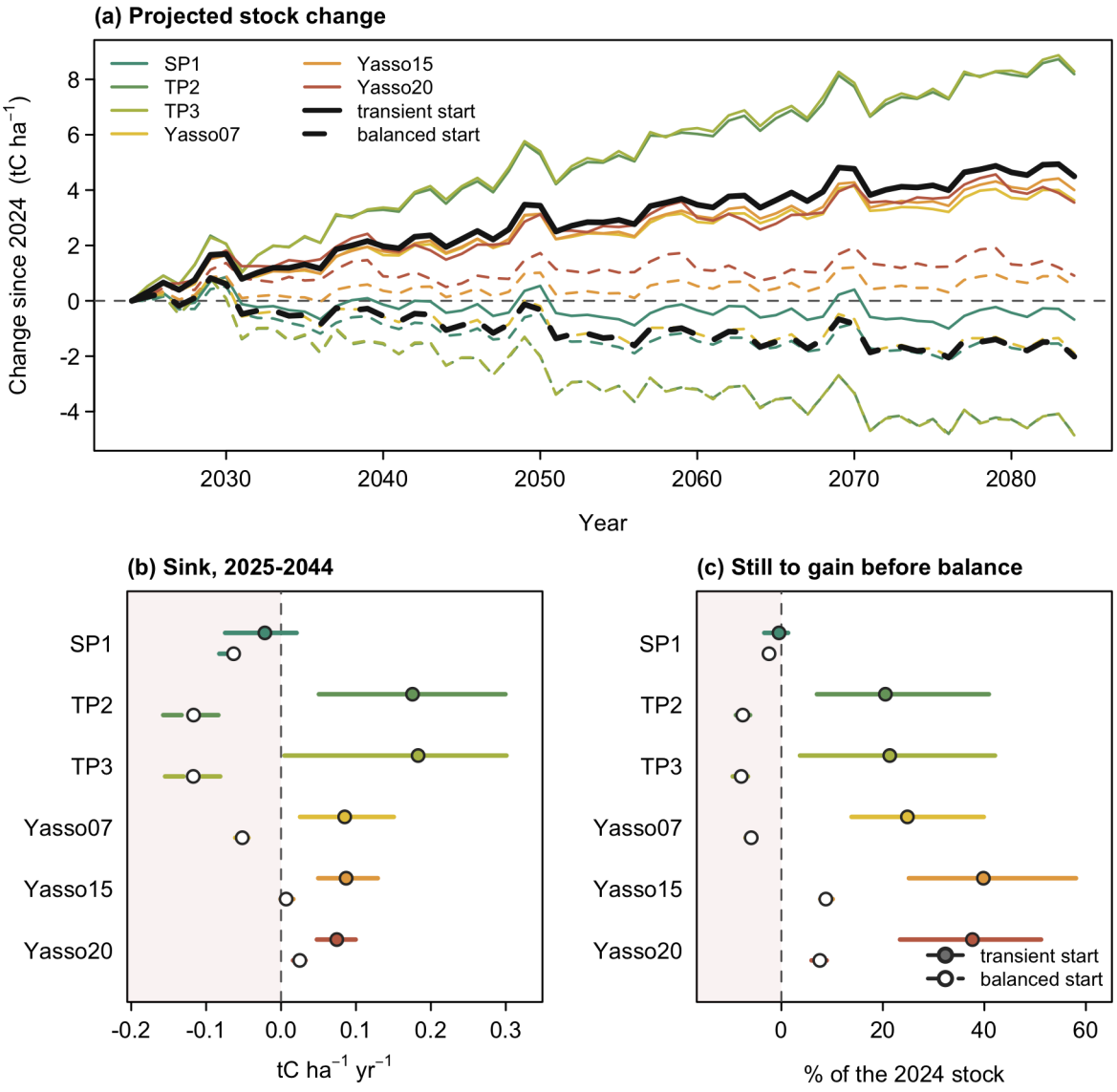


Figure S10: The forecast under the two starting assumptions, with the litter held at 2024 and the climate of 2005–2024 repeated. (a) Projected change of the national mean stock since 2024, transient start (solid) and balanced start (dashed); (b) the sink over 2025–2044; (c) the remaining accumulation to balance, in per cent of the 2024 stock. Bars are 90% intervals over posterior draws of the national mean; plots measured in all three campaigns, unweighted. [Promote?]

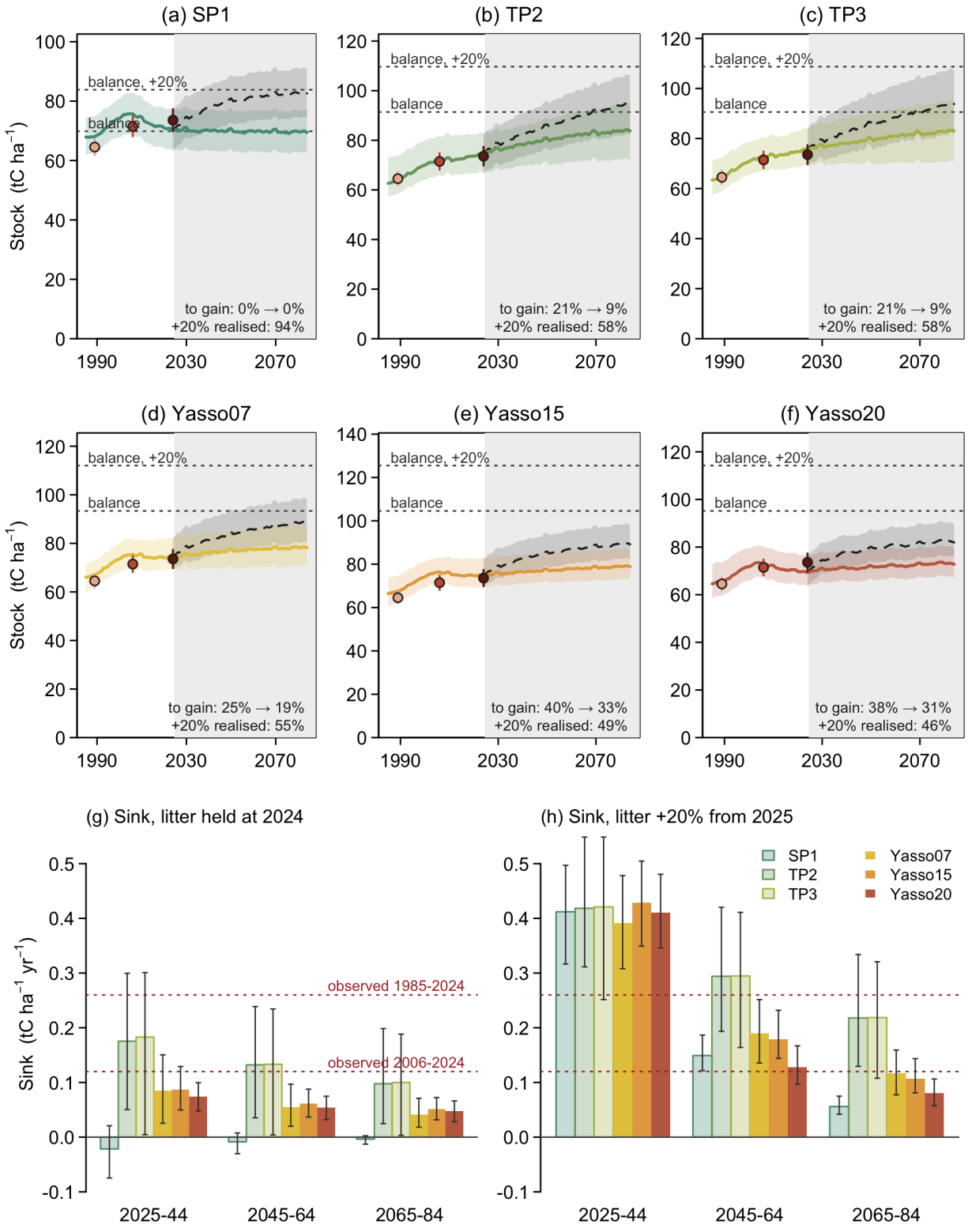


Figure S11: Saturation, postponed but not removed. The projection holds the litter input at its 2024 level and repeats the climate of 2005–2024; the second projection raises the litter input by 20 per cent from 2025 and keeps it there. (a)–(f) National mean stock of each model, 1985–2084: calculated history and projection with the input held (coloured line), projection with the input raised (black dashed line), and the stock at balance under each input (dotted lines); points are the campaign means with 95% intervals, placed at the mean sampling year of each campaign. The text in each panel gives the carbon still to be gained before balance, in per cent of the stock in 2024 and in 2084, and the share of the eventual gain from the raised input that is realised by 2084. (g) The sink with the input held and (h) with the input raised: each bar is the mean sink of one model over one repetition of the twenty-year climate (2025–2044, 2045–2064,

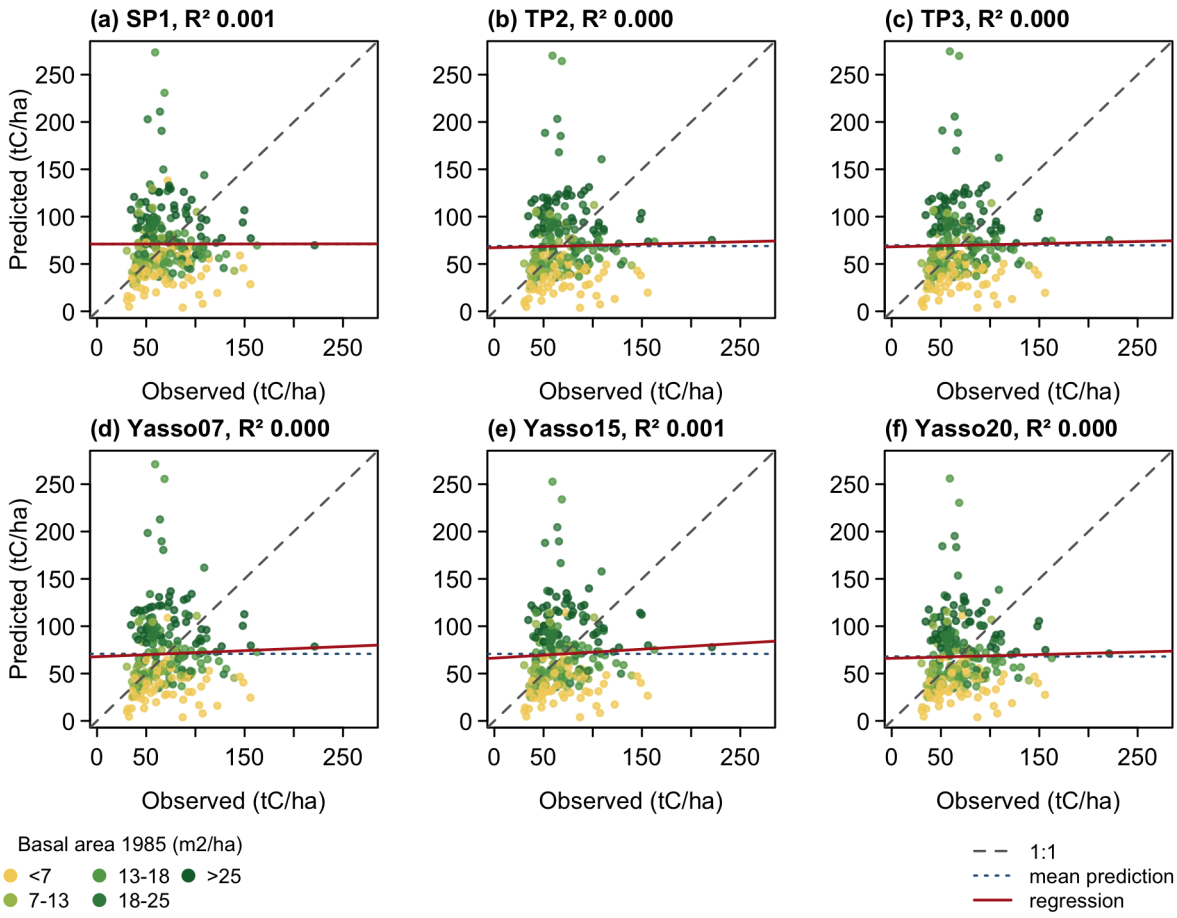


Figure S12: Predicted against measured soil carbon on the held-out plots. Points are coloured by the standing basal area of the plot, in quintiles. The dashed line is the one-to-one line, the dotted line the prediction of the mean, and the solid line the fitted relation. The predicted stocks spread more widely than the measured stocks and are ordered by basal area, whereas the measured stocks are not.

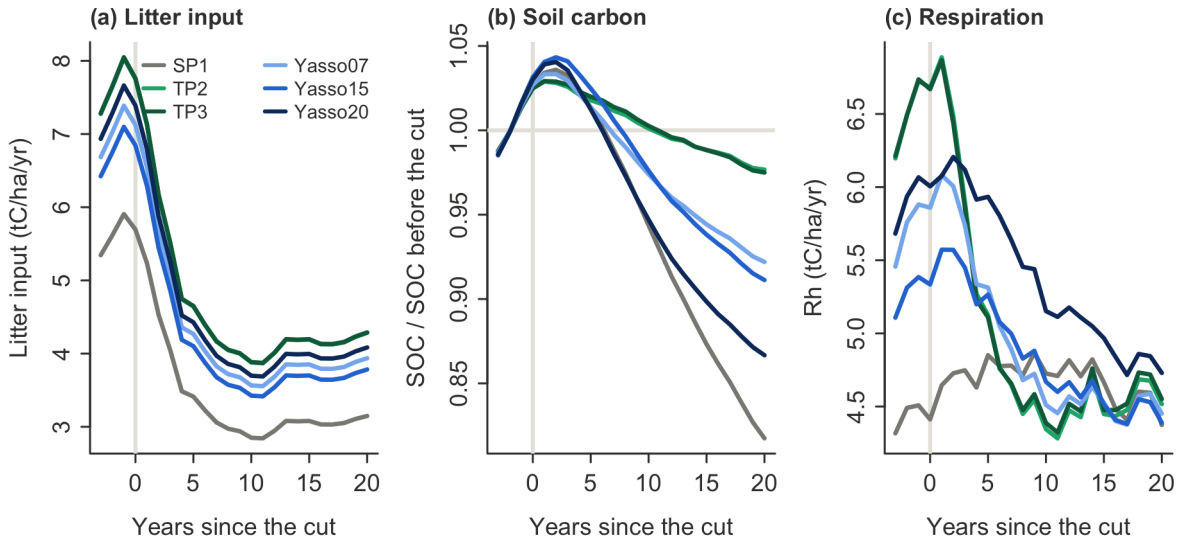


Figure S13: Litter input, soil carbon and heterotrophic respiration around a regeneration cut. Mean over the 63 calibrated plots with a dated regeneration cut in 1989–2004, aligned on the year of the cut, posterior-mean trajectories of the six models. (a) Litter input (the product times the input multiplier); (b) soil carbon relative to the three years before the cut; (c) heterotrophic respiration derived as input minus change of stock (including leaching in Yasso15 and Yasso20). [Keep?]

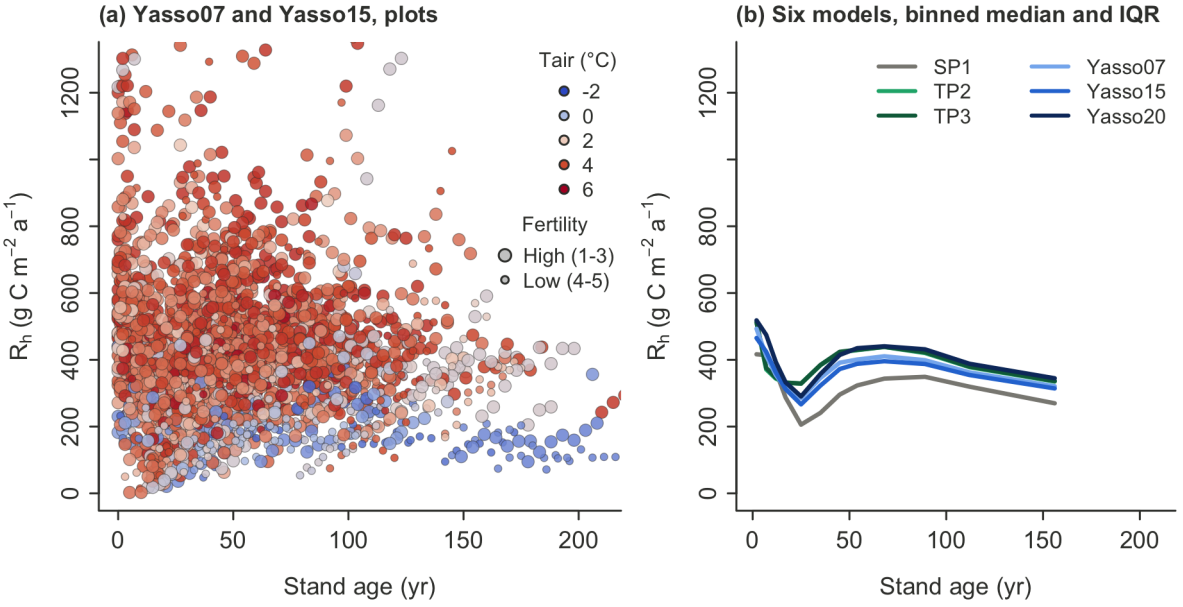


Figure S14: Modelled heterotrophic respiration against stand age. Respiration is the litter input times the posterior median input multiplier minus the annual change of the posterior-mean stock. (a) Mean of Yasso07 and Yasso15, one point per plot every five years over 1990–2020, coloured by air temperature, size by fertility; (b) binned median of the six models. Stand age is the tally age of 1985 plus the years since, reset at dated regeneration cuts after 1985. [Keep?]